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Inventor(s)	Sweeney; Charles A. et al.

Moveable display system

Abstract

Implementations relate to a moveable display system. In some implementations, a control unit includes a first support and a second support coupled to the first support. The second support is linearly translatable along a first axis in a first degree of freedom with respect to the first support, and at least a portion of the second support is linearly translatable along a second axis in a second degree of freedom with respect to the first support. The control unit includes a display unit rotatably coupled to the second support. The display unit is rotatable about a third axis in a third degree of freedom with respect to the second support, and the display unit includes a display device.

Inventors:	**Sweeney; Charles A. (Manteca, CA), Luke; Scott O. (San Jose, CA), Waldo; Michael C. (Los Gatos, CA), Thompson; Allen C. (Los Altos, CA), Goldberg; Randal P. (San Mateo, CA), Lee; Peling G. (Palo Alto, CA), Hingwe; Pushkar (Los Altos, CA), Verner; Lawton N. (San Jose, CA), Lynch; Goran A. (Oakland, CA), Blanchard; Russell L. E. (San Jose, CA)**
Applicant:	**Intuitive Surgical Operations, Inc.** (Sunnyvale, CA)
Family ID:	**1000008821130**
Assignee:	**Intuitive Surgical Operations, Inc. (Sunnyvale, CA)**
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Primary Examiner: Mott; Adam R

Assistant Examiner: Kasper; Byron Xavier

Attorney, Agent or Firm: IP Spring

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a U.S. National Phase application of International Patent Application No. PCT/US2020/047494, filed Aug. 21, 2020 and titled "Moveable Display System," which claims priority to U.S. Provisional Patent Application No. 62/890,844, filed Aug. 23, 2019 and titled "Moveable Display System," the entire contents of both of which are hereby incorporated by reference.

BACKGROUND
(1) Display devices, including display screens, wearable display devices, projectors, etc., are used in a variety of devices and applications. In some applications, a display device outputs images captured by a camera that provide a view of a scene.
(2) In some applications, a display device can be used in conjunction with other devices and allow a user to, e.g., observe a displayed view provided by the display device in conjunction with operating the other devices. For example, in a teleoperated system, a user typically operates a control input device to remotely control (e.g., teleoperate) the motion and/or other functions of a controlled device, such as a manipulator system, at a work site. In some examples, in some teleoperated surgery systems, a user operates a control input device to manipulate surgical instruments and other devices to perform a surgical operation at a surgical site. A control input device often includes hand input devices such as pincher grips, joysticks, exo-skeletal gloves, or the like. In some examples of a surgical or other medical task, a hand input device may control a variety of surgical instruments such as tissue graspers, needle drivers, electrosurgical cautery probes, cameras, etc., which perform functions such as holding or driving a needle, grasping a blood vessel, or dissecting, cauterizing, or coagulating tissue. Other applications can include a variety of telemanipulated tasks performed at a worksite using a teleoperated system.
(3) In various teleoperated systems, a display unit is used in conjunction with the control input device. For example, the display unit can include a display device that displays images depicting a view of a remote work site, or a portion thereof, as captured by a camera at the work site. The display unit can be held by a mechanically-grounded support so that the user can operate the control input device and control a manipulator device to perform tasks at the work site while observing the view of the work site displayed by the display unit. Other systems may also include a display unit that provides a displayed view of a work site to a user without that user using control input devices, e.g., to monitor tasks or events occurring at the work site. In some of these systems, the user can adjust the displayed view by manipulating hand input devices to manipulate a camera, e.g., rotate, pan, and zoom the view of the camera to obtain desired magnification and angles of view of the work site. This allows views of the work site to be customized for more clear presentation, and allows tasks at the work site to be performed accurately based on user-customized views.
(4) However, teleoperated systems may provide a viewing device that has a restrictive viewing area (e.g., the user must peer through viewports or eyepieces) and is static and rigid in its position

relative to the user. The user must conform the user's head and body position to the viewing device during system operation to be able to use the viewing device. For example, a user that wishes to look down to view an object closer to the user must adjust his or her eyes downward without tilting his or her head so that the user can continue to view the images in the restrictive viewing area of the viewing device.

(5) Furthermore, manipulation of the displayed view in a viewing device can be challenging to a user of some systems. For example, if the user is grasping hand input devices of a control input device to move manipulator instruments at a work site, adjusting the view in the viewing device may require the user to provide hand input to a control input device and/or enter a different control mode to change the displayed view. Such operation typically requires the user to interrupt and pause manipulation of manipulator instruments to perform view adjustments, thus causing distractions and potential inaccuracies when performing tasks. Furthermore, some hands-free methods of controlling the displayed view, such as eye tracking sensors or voice commands, are often not precise nor reliable, potentially introducing ambiguity to provided commands from the user.

SUMMARY

(6) Implementations of the present application relate to a moveable display system. In some implementations, a control unit includes a first support and a second support coupled to the first support. The second support is linearly translatable along a first axis in a first degree of freedom with respect to the first support, and at least a portion of the second support is linearly translatable along a second axis in a second degree of freedom with respect to the first support. The control unit includes a display unit rotatably coupled to the second support. The display unit is rotatable about a third axis in a third degree of freedom with respect to the second support, and the display unit includes a display device.

(7) In various implementations of the control unit, the first axis is orthogonal to the second axis, and the third axis is orthogonal to the first axis and orthogonal to the second axis. In some implementations, the first support and the second support are fixed in orientation with respect to each other. In some implementations, the first support includes a first telescoping base portion and a second telescoping base portion, the second telescoping base portion is linearly translatable along the first axis with respect to the first telescoping base portion, the second support includes a first telescoping arm portion and a second telescoping arm portion, the second telescoping arm portion is linearly translatable along the second axis with respect to the first telescoping arm portion, and the second telescoping base portion of the first support is rigidly coupled to the first telescoping arm portion of the second support. In some implementations, the second support is coupled to the first support by a middle support, the middle support includes a horizontal portion coupled rigidly to a vertical portion that are orthogonal to each other, the second support is horizontally translatable in the second degree of freedom with respect to the middle support, and the middle support and the second support are vertically translatable in the first degree of freedom with respect to the first support.

(8) In some implementations, the display unit includes a tilt member rotatably coupled to an end of the second support, the tilt member rotatable about the third axis in the third degree of freedom, and the display unit is coupled to and moveable with respect to the tilt member in a fourth degree of freedom. For example, the display unit can be rotatable about a fourth axis with respect to the tilt member in the fourth degree of freedom, the fourth axis orthogonal to the third axis. In some examples, the control unit further comprises an actuator configured to output a force on the display unit about the fourth axis in the fourth degree of freedom. In some implementations, the portion of the second support includes a yoke portion including two yoke members, and the display unit is rotatably coupled to the two yoke members and is positioned between the two yoke members. In some implementations, the control unit includes a first actuator, a second actuator, and a third actuator, the first actuator configured to output first forces on the display unit in the first degree of

freedom, the second actuator configured to output second forces on the display unit in the second degree of freedom, and the third actuator configured to output third forces on the display unit in the third degree of freedom. For example, the first actuator, the second actuator, and the third actuator can be configured to output the first forces, the second forces, and the third forces in combination to cause rotation of the display unit about a defined pivot axis.

(9) In some implementations, the display unit includes a head input device provided on the display unit and configured to receive input from a head of a user, and/or a hand input device provided on the display unit and configured to receive input from a hand of a user. In some implementations, the control unit is coupled to a device including a control input device manipulable by a user to control of one or more functions of a teleoperated manipulator system.

(10) In some implementations, a teleoperated system control unit includes a vertical member having a first portion and a second portion, the second portion of the vertical member being linearly translatable along a vertical axis with respect to the first portion of the vertical member; a horizontal member having a first portion and a second portion, the first portion of the horizontal member being rigidly coupled to the second portion of the vertical member and the second portion of the horizontal member being linearly translatable along a horizontal axis with respect to the first portion of the horizontal member; and a display unit rotatably coupled to the second portion of the horizontal member and rotatable about a tilt axis with respect to the horizontal member, the display unit including a display device.

(11) In various implementations of the teleoperated system control unit, the first portion of the vertical member and the second portion of the vertical member are telescopically coupled, and the first portion of the horizontal member and the second portion of the horizontal member are telescopically coupled. In some implementations, a tilt member is coupled to the display unit and rotatably coupled to the second portion of the horizontal member, and the tilt member and display unit are rotatable about the tilt axis. In some implementations, the second portion of the horizontal member includes a yoke portion including two parallel members, the display unit being rotatably coupled to, and positioned between, the two parallel members. In some implementations, the teleoperated system control unit further includes a first actuator, a second actuator, a third actuator, and/or a fourth actuator, the first actuator configured to output first forces on the second portion of the vertical member, the second actuator configured to output second forces on the second portion of the horizontal member, the third actuator configured to output third forces on the display unit about the tilt axis, and the fourth actuator configured to output fourth forces on the display unit about a fourth axis in a fourth degree of freedom in which the display unit is rotatable with respect to the tilt member.

(12) In some implementations, the display unit is rotatable about a defined pivot axis, and the rotation of the display unit about the defined pivot axis is caused by a coordinated combination of linear movement of the second portion of the vertical member, linear movement of the second portion of the horizontal member, and rotational movement of the display unit. For example, the defined pivot axis can be positioned at a location such that the defined pivot axis extends through a neck of a user when the user operates the display unit, can extend through a head input device provided on the display unit at a point configured to contact a forehead of the user that operates the head input device, can be coincident with an eye axis extending through eyes of the user that operates the display unit, or can extend through a hand input device provided on the display unit, e.g., the hand input device configured to be operated by a hand of the user when the user operates the display unit. In some implementations, the display unit is rotatable about a yaw axis with respect to the horizontal member in a fourth degree of freedom, and the yaw axis orthogonal to the tilt axis. In some implementations, the teleoperated system control unit is coupled to a device including a control input device manipulable by a user to control of one or more functions of a teleoperated manipulator system.

(13) In some implementations, a control unit includes a support mechanism and a control system.

The support mechanism includes a support linkage that includes a plurality of links, a display unit coupled to the support linkage, and a plurality of actuators coupled to the support linkage. The display unit is moveable in multiple degrees of freedom based on relative movement between the plurality of links, and the display unit includes a display device. The control system is in communication with the support mechanism and is configured to provide control signals to one or more of the plurality of actuators to cause the display unit to rotate about a defined pivot axis. The rotation about the defined pivot axis results from movement of the display unit in at least two of the multiple degrees of freedom.

(14) In various implementations of the control unit, the plurality of links include a tilt member rotatably coupled to a different link of the plurality of links, the tilt member is rotatable about a tilt axis, the display unit is rotatably coupled to the tilt member, and a distance between the defined pivot axis and the tilt axis is fixed during the rotation of the display unit about the defined pivot axis. In various implementations, the multiple degrees of freedom of the support linkage include two or more of a linear first degree of freedom, a linear second degree of freedom, a rotational third degree of freedom, or a rotational fourth degree of freedom. In some examples, the defined pivot axis is: a horizontal axis extending through a neck of a user who operates the display unit, a horizontal axis aligned with an axis extending through eyes of a user who operates the display unit, a horizontal axis aligned with an axis extending through a portion of an input device provided on the display unit, the portion of the input device configured to contact a forehead of a user who operates the display unit, or a horizontal axis aligned with an axis extending through portion of a hand input device provided on the display unit, the hand input device configured to be operated by a hand of a user who operates the display unit and the hand input device. In further examples, the defined pivot axis is adjustable in space based on user input to the control unit.

(15) In some implementations, the control system is configured to provide the control signals to control the actuators to move the display unit about the defined pivot axis to follow movement of a head of a user who operates the control unit. In some example implementations, the display unit is not attached to the head of the user. In some implementations, the control system is in communication with a control input device, and is configured to receive commands from the control input device and send signals based on the commands to a teleoperated manipulator system.

(16) In some implementations, a control unit includes a support, a curved track coupled to the support, and a display unit coupled to the curved track, the display unit including a display device and guided by the curved track to be rotatable about a yaw axis in a rotary degree of freedom. In some implementations, the display unit is movable left or right along the curved track with respect to a user who operates the display unit, the yaw axis extending vertically with respect to the user. In some implementations, the support includes a linkage having a plurality of links moveable with respect to each other, the plurality of links enabling the display unit to move in at least one additional degree of freedom, and an actuator is coupled to the display unit and is configured to output force in the rotary degree of freedom that causes the display unit to move in the rotary degree of freedom. In some examples, the display unit can be rigidly coupled to the curved track and the curved track is slidably coupled to the support, or the display unit can be slidably coupled to the curved track and the curved track is rigidly coupled to the support.

(17) In some implementations, a method includes receiving first user input at a first input device and causing movement of a display unit in one or more degrees of freedom provided by a support linkage coupled to the display unit. The display unit includes a display device. The movement is based on the first user input, and causing the movement includes causing a second link of the support linkage to linearly translate with respect to a first link of the support linkage along a first axis in a first degree of freedom; causing a portion of the second link to linearly translate with respect to the first link along a second axis in a second degree of freedom; and causing the display unit to rotate about a third axis in a third degree of freedom with respect to the second link.

(18) In some implementations, the first user input is received from at least one of a head input

device provided on the display unit that receives input from a head of a user, and/or a hand input device provided on the display unit that receives input from a hand of the user. In some implementations, causing the movement of the display unit includes causing the display unit to rotate about a fourth axis with respect to the second link in a fourth degree of freedom, the fourth axis being orthogonal to the third axis. In some implementations, the method includes receiving second user input at a second input device (e.g., that is coupled to the support linkage), and, based on the second user input, controlling an instrument actuator to move a manipulator instrument in space. For example, the display unit can display a view of a workspace in which the manipulator instrument operates. In some implementations, the method further includes updating images displayed by the display device of the display unit in accordance with the first user input (or otherwise in accordance with the movement of the display unit). In some implementations, the method includes causing a actuator of a manipulator system to move an image capture device of the manipulator system in accordance with the first user input, and image data is received from the image capture device and displayed by the display device of the display unit.

(19) In some implementations, a control unit includes means for receiving first user input at a first input device, means for causing movement, based on the first user input, of a display unit in one or more degrees of freedom provided by a support linkage coupled to the display unit. The display unit includes a display device. The means for causing the movement of the display unit includes means for causing a second link of the support linkage to linearly translate with respect to a first link along a first axis in a first degree of freedom, means for causing a portion of the second link to linearly translate with respect to the first link along a second axis in a second degree of freedom, and means for causing the display unit to rotate about a third axis in a third degree of freedom with respect to the second link. In some implementations, the control unit further comprises means for receiving second user input, and means for controlling a slave instrument actuator, based on the second user input, to move a slave instrument in space.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a diagrammatic view of an example teleoperated surgical system including a control input device, display system, and manipulator device, according to some implementations;

(2) FIG. **2** is a front view of a user control system including control input devices and a display system, according to some implementations;

(3) FIGS. **3-5** are perspective, front, and side views, respectively, of an example display system, according to some implementations;

(4) FIG. **6** is a side view of another example of a portion of display system, according to some implementations;

(5) FIG. **7** is a side view of yet another example of a portion of display system, according to some implementations;

(6) FIGS. **8-11** are side views a portion of a display system showing rotation of a display unit about a defined neck pivot axis, according to some implementations;

(7) FIGS. **12-14** are side views of a portion of display system of FIG. **8** showing rotation of a display unit about a defined eye pivot axis, according to some implementations;

(8) FIGS. **15-16** are perspective and side views, respectively, of an example base support for a display system, according to some implementations;

(9) FIGS. **17** and **18** are perspective and side views, respectively, of an example arm support for a display system, according to some implementations;

(10) FIGS. **19-21** are side, front, and perspective views, respectively, of an example tilt mechanism, according to some implementations;

(11) FIG. **22** is a perspective view of an example display unit mechanism, according to some implementations;
(12) FIGS. **23-25** are perspective, front, and side views, respectively, of another implementation of a display system, according to some implementations;
(13) FIGS. **26-29** are side views of a portion of a display system showing rotation of a display unit about a defined neck pivot axis, according to some implementations;
(14) FIGS. **30** and **31** are side views of a portion of display system of FIG. **26** showing rotation of a display unit about a defined eye pivot axis, according to some implementations;
(15) FIG. **32** is a perspective view of another implementation of a display system, according to some implementations;
(16) FIGS. **33-36** are front, side, top, and bottom views, respectively, of a portion of the display system of FIG. **32**, according to some implementations;
(17) FIGS. **37** and **38** are perspective views of a portion of the display system of FIG. **32**, according to some implementations;
(18) FIG. **39** is a side view of a portion of the display system of FIG. **32** showing defined pivot axes, according to some implementations;
(19) FIG. **40** is a perspective view of an example portion of a control input device that can be used with the described display systems, according to some implementations;
(20) FIG. **41** is a flow diagram illustrating an example method for operating a display system including one or more features described herein, according to some implementations; and
(21) FIG. **42** is a block diagram of an example master-slave system, which can be used with one or more implementations described herein.

DETAILED DESCRIPTION

(22) Implementations relate to a moveable display system that accommodates and/or is responsive to user viewing. For example, in some implementations, the display system is used in a user control system to provide a displayed view of a work site or environment to a user while the user operates one or more other devices, such as a control input device of a teleoperated system. As described in more detail herein, implementations provide a display system that is responsive to the user's motions to move a viewed display unit in ways that accommodate and/or follow user head motion. In various examples, the display system can include a display unit that displays images and is mechanically grounded, allowing the user to manipulate other devices, such as hand-operated and/or foot-operated control input devices, while viewing the output of the display unit. Some implementations can provide various views of a work site by the display unit via camera images that are responsive to user input provided via the display system.
(23) Described features of the display system include a support linkage providing multiple degrees of freedom to the display unit. The support linkage includes a first (e.g., vertical) support having a first portion and a second portion, and the second portion is linearly translatable along a vertical axis with respect to the first portion (e.g., telescoping portions). The support linkage includes a second (e.g., horizontal) support rigidly coupled to the vertical support and having a first portion and a second portion, and the first portion is rigidly coupled to the second portion of the vertical member and the second portion is linearly translatable along a horizontal axis with respect to the first portion (e.g., telescoping portions). In some implementations, the support linkage includes a tilt member rotatably coupled to a distal end of the second support, the tilt member rotatable about a third axis in a third degree of freedom. In some implementations, a display unit is coupled to the tilt member. In some implementations, the display unit is moveable with respect to the tilt member in a fourth degree of freedom, e.g., rotational movement about a fourth (yaw) axis. Actuators such as motors are coupled to one or more of these components to allow a control system to move the components and thereby move the display unit in a workspace in particular degrees of freedom. User input devices on the display unit allow the user to provide user input to direct the movement, e.g., change in position and orientation, of the display unit. The display system can be used in

conjunction with a control input device that provides control signals to a manipulator system in a teleoperated system to control manipulator system functions, e.g., movement and other functions of manipulator instruments.
(24) Described features provide various benefits. For example, the support linkage and actuators allow the display system to change the position and orientation of the display unit based on received user input, e.g., to accommodate changes in angle and motion of the user's head and eyes or to respond to commands input by the user's hands, head, eyes, etc. For example, the display unit can be rotated about a defined pivot axis that can coincide with the user's eyes or coincide with a pivot axis of the user's neck, thus providing motion of the display unit that is aligned with the user's natural body motions. These features allow the user to easily re-orient the display unit during procedures to obtain different comfortable viewing angles and to reduce the physical constraints of using the display unit, thus decreasing fatigue of the user in associated procedures. The motion of display unit in the provided tilt, horizontal, and vertical degrees of freedom, e.g., providing motion about a defined pivot axis and a yaw axis, allows the display unit to follow and stay close to the user's head and/or eyes (e.g., movement of the display unit mirrors or copies movement of the user's head and/or eyes) during user head motion, and/or to maintain a physical connection between the user's forehead and the display unit. In some implementations, this allows the display unit to follow movement of a user's head and eyes without having to physically attach the display unit to the user's head, thus avoiding user irritation and fatigue from such attachment.
(25) Furthermore, the defined pivot axis of the display unit can be a virtual axis that need not be confined to physical axes of motion of the mechanical components of the display system. This allows the location of the defined pivot axis to be adjusted for various use conditions and/or customized for a particular user, e.g., to accommodate a particular user's height, arm reach, size of neck or head, etc., to allow greater comfort in operating the display system. Furthermore, sensors such as head sensors on the display unit allow the user to easily provide user input without using hands that may be manipulating other input devices such as a control input device.
(26) In addition, changes in the position and/or orientation of the display unit, directed by user input, can be used to modify the display of images by the display unit. For example, a displayed image or user interface can be scrolled, tilted, panned, or zoomed based on corresponding received user input to the display device that also direct the display unit to perform similar or corresponding motions. In some implementations, functions of an image capture device (or other instrument or device) at a remote work site are manipulated based on the user input to the display unit, e.g., movement of the image capture device or other device functions such as panning, tilting, and zooming. Head sensors on the display unit allow the user to provide such user input without having to interrupt or pause a teleoperated procedure using other user input devices such as a control input device operated by hand or foot.
(27) Various terms including "linear," "center," "parallel," "orthogonal," "perpendicular," "aligned," "horizontal," "vertical," or particular measurements or other units as used herein can be approximate, need not be exact, and can include typical engineering tolerances.
(28) Some implementations herein may relate to various instruments and portions of instruments in terms of their state in three-dimensional space. As used herein, the term "position" refers to the location of an object or a portion of an object in a three dimensional space (e.g., three degrees of translational freedom along Cartesian X, Y, Z coordinates). As used herein, the term "orientation" refers to the rotational placement of an object or a portion of an object (three degrees of rotational freedom—e.g., roll, pitch, and yaw around the Cartesian X, Y, and Z axes). As used herein, the term "pose" refers to the position of an object or a portion of an object in at least one degree of translational freedom and to the orientation of that object or portion of the object in at least one degree of rotational freedom (up to six total degrees of freedom).
(29) As referred to herein, a mechanically grounded unit or device is constrained with respect to possible position and orientation motion in a large working environment (e.g., an operating area or

room). Also, such a unit is kinematically coupled to the ground (e.g., mechanically supported by a console, supports, or other object attached to the ground). As used herein, the term “proximal” refers to an element that is close to (or closer to) a mechanical ground and the term “distal” refers to an element that is away from (or further from) a mechanical ground.

(30) Various features described herein can be used to augment the control capability of a computer-assisted teleoperated system. In some implementations, the teleoperated system includes one or more control input devices (e.g., one, two, three, or more) for providing manipulator instrument control in various procedures (surgical, procedures in extreme environments, or other procedures), instruction, supervision, proctoring, and other feedback to a user of the system.

(31) FIG. **1** is a diagrammatic view of an example teleoperated surgical system **100**, which can be used with one or more features disclosed herein. As shown, teleoperated surgical system **100** may include a user control system (e.g., console or workstation) **102** and a manipulator system **104**.

(32) In this example, the user control system **102** includes one or more control input devices which are contacted and manipulated by the user's hands, e.g., one control input device for each hand. FIGS. **2** and **40** show some example implementations of control input devices which are described in greater detail below. The control input devices are supported by the user control system **102** and can be mechanically grounded. An ergonomic support **110** (e.g., forearm rest) can be provided in some implementations, on which user **108** can rest his or her forearms while grasping control input devices. For example, the control input devices can be positioned in a workspace disposed inwardly (away from user **108**) beyond the support **110**. In some examples, the user **108** may perform surgical tasks at a work site near the manipulator system **104** during a surgical procedure by controlling the manipulator system **104** using the control input devices.

(33) A display unit **112** is included in the user control system **102**. Display unit **112** can display images for viewing by the user **108**. For example, the images can be displayed by a display device in the display unit, such as one or more display screens, projectors, or other devices. The display unit **112** can be moved in various degrees of freedom to accommodate the user's viewing position and/or to provide control functions, as described in greater detail below. In the example of the teleoperated system **100**, displayed images can depict a work site at which the user is performing various tasks via control of the control input devices. In some examples, the images displayed by the display unit **112** can be received by the user control system **102** from one or more image capture devices arranged at a remote work site. In other examples, the images displayed by the display unit can be generated by the display unit (or by a connected other device or system). In an example of a surgical procedure using teleoperated system **100**, the display unit **112** can display images of a physical surgical site at a patient near the manipulator system **104**, or a generated virtual representation of a surgical site or a combination of physical and virtual sites (e.g., augmented reality), and can display real or virtual instruments of the manipulator system **104** controlled by the control input devices of user control system **102**. Display unit **112** can provide a two dimensional image and/or a three-dimensional image of, for example, an end effector of a manipulator instrument **126** and the surgical site. A three-dimensional image can provide three-dimensional depth cues to permit user **108** to assess relative depths of instruments and patient anatomy and to use visual feedback to steer the manipulator instruments **126** using control input devices to precisely target and control features.

(34) When using the user control system **102**, the user **108** can sit in a chair or other support in front of the user control system **102**, position his or her eyes in front of the display unit **112** (and/or move the display unit **112** to a position/orientation of his or her eyes), grasp and manipulate the control input devices, e.g., one in each hand, and rest his or her forearms on the ergonomic support **110** as desired. In some implementations, the user can stand at the user control system or assume other poses, and the display unit **112** and control input devices can be adjusted in position (height, depth, etc.) to accommodate various user body poses and individual user preferences.

(35) The teleoperated system **100** may also include manipulator system **104** which can be

controlled by the user control system **102**. In this example, the manipulator system **104** is mounted to or near an operating table **106** (e.g., table, bed, or other support) on which a patient may be positioned. A work site **130** can be provided on the operating table **106**, e.g., on or in a patient, simulated patient or model, etc. (not shown). In other implementations, a work site can be a different site or area at which tasks are to be performed using a manipulator system. The teleoperated manipulator system **104** includes a plurality of manipulator arms **120**, each coupled to an instrument assembly **122**. An instrument assembly **122** may include, for example, an instrument **126**. In some examples, instruments **126** may include surgical instruments. In some implementations, a surgical instrument can include a surgical end effector at its distal end, e.g., for treating tissue of a patient.

(36) In various implementations, one or more of the instruments **126** can include image capture devices (e.g., cameras), such as a camera included in an endoscope assembly **124**, which can provide captured images of a portion of the work site (e.g., a region or portion of a patient in which a surgical task is being performed). In some implementations, captured images can be transmitted to the display unit **112** of the user control system **102** for output. In some implementations, a display device **128** can be included on the manipulator system **104** to display captured images and/or other information related to a procedure being performed at the work site. In some implementations, an image capture device can be moved in multiple degrees of freedom, e.g., based on translation and rotation of portions of a manipulator arm **120** holding the camera.

(37) In an example of a surgical procedure using the teleoperated system **100**, the manipulator system **104** can be positioned close to a patient (or simulated patient) for surgery, where it can remain stationary until a particular surgical procedure or stage of a procedure is completed. In various implementations, the user control system **102** can be positioned in various locations relative to the manipulator system **104**, e.g., in a sterile surgical field close to manipulator system **104** and the work site **130**, in the same room as the manipulator system **104** and work site **130**, or remotely from the manipulator system **104** and work site **130**, e.g., in a different room, building, or other geographic location. The number of teleoperated instruments **126** used at one time, and/or the number of arms **120** used in manipulator system **104**, may depend on the procedure to be performed and the space constraints within the available area, among other factors.

(38) In some implementations, the manipulator arms **120** and/or instrument assemblies **122** may be controlled to move and articulate the instruments **126** in response to manipulation of control input devices by the user **108**, so that the user **108** can perform tasks at the work site **130**. For example, the user can direct surgical procedures at internal surgical sites through minimally invasive surgical apertures. In some implementations, one or more actuators coupled to the manipulator arms **120** and/or instrument assemblies **122** may output force to cause links or other portions of the arms **120** and/or instruments **126** to move in particular degrees of freedom in response to control signals received from the control input devices.

(39) Some implementations of the teleoperated system **100** can provide different modes of operation. In some examples, in a non-controlling mode (e.g., safe mode) of the teleoperated system **100**, the controlled motion of the manipulator system **104** is controllably decoupled (disconnected) from the control input devices in disconnected configuration, such that movement and other manipulation of the control input devices do not cause motion of the manipulator system **104**. In a controlling mode of the teleoperated system **100** (e.g., following mode), motion of the manipulator system **104** can be controllably coupled (connected) to the control input devices such that movement and other manipulation of the control input devices causes motion of the manipulator system **104**, e.g., during a surgical procedure. For example, each manipulator arm **120** and the teleoperated instrument assembly **122** controlled by that arm **120** may be controllably coupled to and decoupled from one or more control input devices to allow control over movement and/or other functions of that arm.

(40) In some examples, the control over manipulator systems enables the user to direct surgical

procedures at internal surgical sites through minimally invasive surgical apertures. For example, one or more actuators coupled to the manipulator arms **120** can output force to cause links or other portions of the arms to move in particular degrees of freedom in response to control signals provided by the control input devices. The control input devices can be used within a room (e.g., an operating room) that also houses the manipulator system and worksite (e.g., within or outside a sterile surgical field close to an operating table), or can be positioned more remotely from the manipulator system, e.g., at a different room, building, or other location than the manipulator system.

(41) In some implementations, a control system (not shown in FIG. **1**) is provided in user control system **102** and/or is provided externally to the user control system **102** (e.g., in communication with the user control system). As the user **108** moves control input device(s), sensed spatial information and sensed orientation information is provided to the control system based on the movement of the control input devices. Other user input is also provided to the control system, e.g., user input received at the display unit **112**, and/or activation of other input devices. The control system can provide control signals to the manipulator system **104** to control the movement of arms **120**, instrument assemblies **122**, and instruments **126** based on the received information and user input. For example, the control system can map sensed spatial motion data and sensed orientation data describing the control input devices in space to a common reference frame. The control system may process the mapped data and generate commands to appropriately position an instrument **126**, e.g., an end effector or tip, of manipulator system **104** based on the movement (e.g., change of position and/or orientation) of one or more control input devices. The control system can use a teleoperation servo control system to translate and to transfer the sensed motion of the control input devices to an associated arm **120** of the manipulator system **104** through control commands so that user **108** can manipulate the instruments **126** of the manipulator system **104**. The control system can similarly generate commands based on activation or manipulation of input controls of the control input devices to perform other functions of the manipulator system **104** and/or instruments **126**, e.g., move jaws of an instrument end effector, activate a cutting tool or output energy, activate a suction or irrigation function, etc. In some implementations, the control system can similarly generate commands based on activation or manipulation of input controls of the display unit **112** to perform other functions of the manipulator system **104** and/or instruments **126**. In one embodiment, the control system supports one or more wireless communication protocols such as Bluetooth, IrDA, HomeRF, IEEE 802.11, DECT, and Wireless Telemetry. Some examples of a control system are described below with respect to FIG. **34**.

(42) In some implementations, the display unit **112** can be operated by a user in conjunction with the operation of one or more ungrounded control input devices, which are control input devices that are not kinematically grounded, e.g., control input devices held by the user's hands without additional support. For example, the user can sit or stand and view images in display unit **112** while grasping and manipulating ungrounded control input devices in his or her hands. Some examples of an ungrounded control input device are disclosed in U.S. Pat. No. 8,521,331 B2 (issued on Aug. 27, 2013, titled “Patient-side Surgeon Interface For a Minimally Invasive, Teleoperated Surgical Instrument”), which is incorporated herein by reference in its entirety. In some implementations, the user can use display unit **112** that is positioned near to the work site such that the user can operate manual surgical instruments at the work site, such as a laparoscopic instrument or a stapler, while viewing images displayed by the display unit **112**.

(43) In some implementations, the teleoperated system **100** may also include one or more additional input systems which allow additional users to provide input to the system. For example, a second (or third, etc.) display unit **112** can be used by an additional user to monitor and/or assist the procedure. A second user control system **102** can be provided for use by a second user, e.g., for training, to alternate control or provide simultaneous control of the manipulator system **104**, etc. Additional ungrounded control input devices, side carts with display devices, and other

components can be used in the teleoperated system **100**.

(44) In some implementations, a virtual representation of manipulator system **104** can be controlled instead of the physical manipulator system **104**, e.g., presented in a graphical training simulation provided by a computing device coupled to the teleoperated system **100**. For example, a user can manipulate control input devices to control a displayed representation of an end effector in virtual space of the simulation, similarly as if the end effector were a physical object coupled to a physical manipulator system. Some implementations can use control input devices in training, e.g., demonstrate the use of instruments and controls of a user control system including control input devices.

(45) In some implementations, non-teleoperated systems can also use one or more features of the user control system and/or display unit **112** as described herein. For example, various types of control systems and devices, peripherals, etc. can be used with described display unit systems. In some examples, display unit **112** can be used in some non-teleoperated systems, e.g., to view a remote work site or physical scene at which the user does not manipulate a manipulator system, to view a displayed virtual environment unrelated to a physical manipulator system or physical work site, etc. In some of these systems, the user control system **102** and manipulator system **104** can be omitted and the display unit **112** can be used in a standalone display system.

(46) Some implementations can include one or more components of a teleoperated medical system such as a da Vinci® Surgical System (e.g., a Model IS3000 or IS4000, marketed as the da Vinci® Si® or da Vinci® Xi® Surgical System), commercialized by Intuitive Surgical, Inc. of Sunnyvale, California. Features disclosed herein may be implemented in various ways, including in implementations at least partially computer-controlled, controlled via electronic control signals, manually controlled via direct physical manipulation, etc. Implementations on da Vinci® Surgical Systems are merely examples and are not to be considered as limiting the scope of the features disclosed herein. For example, different types of teleoperated systems having manipulator systems at work sites can make use of features described herein. Other, non-teleoperated systems can also use one or more described features, e.g., various types of control systems and devices, peripherals, etc.

(47) FIG. **2** is a front elevational view of an example user control system **200** including control input devices and a display system, according to some implementations. For example, user control system **200** can be similar to the user control system **102** described for FIG. **1**.

(48) User control system **200** includes display unit **204** which can, for example, display images during a procedure implemented by the teleoperated system **100** similarly as described for display unit **112** of FIG. **1**. The images can be captured by an image capture device and depict a physical work site at which a task is performed, such as a surgical site displayed during a surgical procedure, or can depict a generated representation of a virtual work site. The display unit **204** can also display other information, such as a graphical user interface allowing selection of commands and functions, status information, alerts and warnings, notifications, etc. Such information can be displayed in combination with (e.g., overlaid on) a view of a work site, or without a work site view.

(49) In the example shown, the display unit **204** includes two viewports **205**. A user can position the user's head such that the user's eyes are aligned with the viewports **205** to view images displayed by the display unit **204**. Furthermore, as described herein, display unit **204** is moveable (translatable and/or rotatable) within a defined workspace based on user input such that the user can align the viewports of the display unit and a viewing angle of the display unit with the user's eyes. In some examples, one or more display screens can be provided behind the viewports which display images to the viewing user. In some implementations, one or more display screens or other display devices can be used instead of viewports **205**. The display unit **204** is connected to a support mechanism and can be moved in one or more degrees of freedom, examples of which are described in greater detail below.

(50) In some implementations, the user control system **200** includes one or more input control

devices to allow a user to adjust or otherwise manipulate the position and/or orientation of the display unit **204** with respect to the other portions of the user control system **200** (e.g., with respect to control input devices **210** and **212**, described below). In this example, a hand input device **206***a* is positioned on the left side of the display unit **204** and a hand input device **206***b* on the right side of the display unit **204**. In some implementations, the hand input devices **206***a* and **206***b* can receive user input to cause the display unit **204** to change its orientation and/or position, e.g., to provide ergonomic adjustments for more user comfort. Such hand input devices can alternatively or additionally be positioned at other areas or components of the user control system **200**. Examples of hand input devices **206***a* and **206***b* are described in greater detail below.

(51) In some implementations, a head input device **208** is positioned on a side of the display unit **204** that is facing the user. Head input device **208** can sense a user's head (e.g., forehead), e.g., sense presence and/or contact with the user's head, as user input to cause the display unit **204** to be moved in accordance with the user input, e.g., change the orientation and/or position of the display unit **204**. Additionally or alternatively, the control system can command one or more components, changes in state, or processes of the user control system **102** and/or manipulator system **104** in accordance with the user input to head input device **208**. Examples of head input device **208** are described in greater detail below.

(52) Two control input devices **210** and **212** are provided in user control system **200** for user manipulation. For example, a user can rest the user's forearms on an ergonomic support **214** while gripping portions of the two control input devices **210** and **212**, with one control input device in each hand (e.g., the control input devices **210** and **212** can be moved to positions above the support **214** and/or the support **214** can be moved to positions lower than the control input devices **210** and **212**). In some implementations, ergonomic support **214** can be adjustable in height for different users. The user also positions the user's head to view the display unit **204** as described above while manipulating the control input devices **210** and **212**. The control input devices may include one or more of any number of a variety of input devices manipulable by the user, such as kinematically linked (mechanically grounded) hand grips, joysticks, trackballs, data gloves, trigger-guns, hand-operated controllers, voice recognition devices, touch screens, and the like.

(53) In some implementations, each control input device **210** and **212** can include grip portions that are moveable in a plurality of degrees of freedom. For example, each control input device **210** and **212** can control motion and functions of an associated arm assembly **120** of the manipulator system **104** shown in FIG. **1**. In some examples, control input device **210** can be moved in a plurality of degrees of freedom to move one corresponding end effector **126** of the manipulator system **104** in corresponding degrees of freedom, and control input device **212** can be moved in a plurality of degrees of freedom to move a different corresponding end effector **126** of the manipulator system **104** in corresponding degrees of freedom. In some implementations, the control input devices **210** and **212** are provided with the same degrees of freedom as the instruments **126** of the manipulator system to provide the operator with telepresence, e.g., the perception that the control input devices are integral with the instruments so that the operator has a strong sense of directly controlling instruments as if present at the work site. In other implementations, the control input devices **210** and **212** may have more or fewer degrees of freedom than the associated instruments **126**. In some implementations, the control input devices are manual input devices which move in all six Cartesian degrees of freedom, and which may also include an actuatable grip portion (e.g., handle) for actuating manipulator instruments, e.g., for closing grasping jaws, applying an electrical potential to an electrode, delivering a medicinal treatment, and the like. In some implementations, a grip function, such as moving two grip portions of a control input device together and apart in a pincher movement, can provide an additional mechanical degree of freedom (i.e., a grip DOF). In some example implementations, control input devices **210** and **212** may provide control of one or more surgical instruments **126** in a surgical environment or proxy surgical instruments in a virtual environment. Examples of some implementations of a control input device are described with

respect to FIG. **32**.

(54) Some implementations of user control system **200** can include one or more foot controls **220** positioned below the control input devices **210** and **212**. The foot controls **220** can be depressed, slid, and/or otherwise manipulated by a user's feet to input various commands to a control system of a teleoperated system while the user is operating the user control system **200**. In some implementations, foot controls **220** or other controls can be considered “control input devices” that can control one or more functions or operations of the manipulator system **104**.

(55) In some implementations, one or more user presence sensors can be positioned at one or more locations of the user control system **200** to detect the presence of a user operating the user control system **200** and/or located next to or near to the user control system **200**. For example, user presence sensors can be positioned on display unit **204** and sense a presence of a user's head aligned with the viewports **205**. For example, an optical sensor can be used for a presence sensor, where the optical sensor includes an emitter and a detector and an interruption of an optical beam is sensed by the detector when the user's head is positioned to view the output of the display unit **204** and the user is in a proper position to use the control input devices **210** and **212**. In some implementations, hand input devices **206** and/or head input device **208** can be used to sense user presence. Additional or alternative types of presence sensors can be used in various implementations.

(56) FIG. **3** is a perspective view, FIG. **4** is a front view, and FIG. **5** is a side view of an example display system **300**, according to some implementations. In some examples, display system **300** can be used in a user control system of a teleoperated system, e.g., user control system **102** of teleoperated surgical system **100** of FIG. **1**, or can be used in other systems or as a standalone system, e.g., to allow a user to view a work site or other physical site, a displayed virtual environment, etc.

(57) Display system **300** includes a base support **302**, an arm support **304**, and a display unit **306**. As described in greater detail below, display unit **306** is provided with multiple degrees of freedom of movement provided by a support linkage including base support **302**, arm support **304** coupled to the base support **302**, and a tilt member **324** (described below) coupled to the arm support **304**. Display unit **306** is coupled to the tilt member.

(58) Base support **302**, in this example, is a vertical member that is mechanically grounded, e.g., coupled to ground. For example, base support **302** can be mechanically coupled to a support structure **310** that is coupled to (e.g., resting on) the ground to provide stability to the base support **302**. Base support **302** includes a first base portion **312** and a second base portion **314**. The first base portion **312** is a proximal portion of the base support **302** that can be mechanically grounded, and the second base portion **314** is a distal portion of the base support **302** that is linearly coupled to the first base portion **312** such that the second base portion **314** is translatable with respect to the first base portion **312** in a linear degree of freedom. In some examples, first base portion **312** and second arm portion **314** are telescopically coupled, e.g., first base portion **312** is a first telescoping base portion and second base portion **314** is a second telescoping base portion, such that one of the portions **312** or **314** is configured as a tube or sleeve with a hollow interior through which the other of the portions **314** or **312** extends. In the example of FIGS. **3-5**, second base portion **314** is linearly translatable through an interior of first base portion **312** in a linear degree of freedom **316**. The linear translation of second base portion **314** with respect to first base portion **312** can be driven by one or more actuators, e.g., motors, as described in greater detail below. Other implementations can use different configurations. For example, first base portion **312** can extend through the interior of second base portion **314** such that second base portion **314** can be linearly translated with respect to first base portion **312**. In other examples, the base portions **312** and **314** can be positioned adjacent to each other along their vertical lengths to allow the linear translation.

(59) Arm support **304** is a horizontal member that is mechanically coupled to the base support **302**. Arm support **304** can include a first arm portion **318** and a second arm portion **320**. The first arm

portion **318** is a proximal portion of the arm support **304** that is rigidly coupled to the second base portion **314** of base support **302**, and the second arm portion **320** is a distal portion of the arm support **304** that is linearly coupled to the first arm portion **318** such that the second arm portion **320** is linearly translatable with respect to the first arm portion **318** in a linear degree of freedom. In some examples, first arm portion **318** and second arm portion **320** are telescopically coupled, e.g., first arm portion **318** is a first telescoping arm portion and second arm portion **320** is a second telescoping arm portion, such that one of the portions **318** or **320** is configured as a tube or sleeve with a hollow interior through which the other of the portions **320** or **318** extends. In the example of FIGS. **3-5**, second arm portion **320** is linearly translatable through an interior of first arm portion **318** in a linear degree of freedom **322**. The linear translation of second arm portion **320** with respect to first arm portion **318** can be driven by one or more actuators, e.g., motors, as described in greater detail below. Other implementations can use different configurations, e.g., first arm portion **318** can extend through the interior of second arm portion **320** such that second arm portion **320** can be linearly translated with respect to first arm portion **318**. In other examples, the arm portions **318** and **320** can be positioned adjacent to each other along their vertical lengths to allow the linear translation.

(60) In some implementations, first arm portion **318** and second base portion **314** can be considered to be a single piece, e.g., a middle support or middle portion that is coupled between first base portion **312** and second arm portion **320**. The middle support includes the horizontal first arm portion **318** coupled rigidly to the vertical second base portion **314** that are oriented orthogonally to each other. Second arm portion **320** is horizontally translatable in the degree of freedom **322** with respect to the middle support, and the middle support and second arm portion **320** are vertically translatable in the degree of freedom **316** with respect to first base portion **312**.

(61) In some examples as shown, arm support **304** extends along a horizontal axis that is orthogonal to a vertical axis along which base support **302** extends. In some examples, base support **302** and arm support **304** are fixed in orientation with respect to each other, e.g., they translate but do not change orientation with respect to each other. In some examples, the arm support **304** extends along an axis above a user operating the display unit, and a vertical axis extending through the base support **302** extends through the first arm portion **318** of the arm support **302**. In other implementations, arm support **304** can extend at other heights and/or configurations, e.g., below a user's head or body, at the height of the user's head, in back of user and yoking around a user, etc.

(62) Reducing vibration in the supports and members of the display system **300** can cause a smoother experience for a user operating the display unit **306**. Some implementations can include one or more components in base support **302** and/or arm support **304** to provide damping that can reduce vibration in the support linkage that includes these supports **302** and **304**. For example, a portion of a lever arm can be rigidly coupled internally to arm support **304** that brushes or compresses a highly damped material provided in the base support **302** to reduce vibration in the arm support **304**. For example, Sorbothane® material, a viscous material, or other material having a damping coefficient above a particular threshold can be used. Similar damping components can be provided between the display unit **306** and tilt member **324**, between tilt member **324** and arm support **302**, etc.

(63) Display unit **306** is mechanically coupled to arm support **304**. Display unit **306** is moveable in two linear degrees of freedom provided by the linear translation of the second base portion **314** and second arm portion **320**. In some implementations, these linear degrees of freedom can be provided within a vertical plane. In some examples, as shown, the vertical plane can be defined by the base support **302** and arm support **304**.

(64) Display unit **306** includes a display device, e.g., one or more display screens, projectors, or other display devices, that can display digital images. In some implementations, as in FIGS. **3-5**, the display unit **306** includes two viewports **323**, and the display device is provided behind or included in the viewports. In some implementations, one or more display screens or other display

devices can be positioned on the display unit **306** in place of viewports **323**.

(65) Display unit **306** is rotationally coupled to the arm support **304** by a tilt member **324**. In the example of FIGS. **3-5**, tilt member **324** is rotationally coupled at a first end to the second arm portion **320** of the arm support **304** by a rotary coupling providing rotational motion of the tilt member **324** and display unit **306** about tilt axis **326** with respect to the second arm portion **320**. In some implementations, tilt axis **326** is oriented orthogonally to the linear degrees of freedom provided to the display unit **306** by base support **302** and arm support **304**. For example, tilt member **324** can provide a rotary degree of freedom to display unit **306** that is in a vertical plane the same as, or is parallel to, the vertical plane in which the degrees of freedom **316** and **322** are provided by base support **302** and arm support **304**. In some implementations, tilt axis **326** is orthogonal to the plane defined by the degrees of freedom **316** and **322**. In some implementations, tilt axis **326** is positioned above the display device in the display unit **306**. In some implementations, tilt axis **326** is positioned above a position of a user's head when the user operates the display unit **306**. In other implementations, the tilt axis can be positioned closer to the user, e.g., lower and closer to a pivot axis of the user's neck as described below. In some examples, arm support **304** can have two branches that extend on either side of the user's head, where the tilt axis **326** extends between the ends of the two branches and is aligned with a defined pivot point in the user's head or neck.

(66) In this example, an extended portion of tilt member **324** extends toward the base support **302** from tilt axis **326** and display unit **306** is coupled to a second end of the tilt member at the extended portion. The rotational motion of tilt member **324** and display unit **306** about tilt axis **326** can be driven by one or more actuators, e.g., motors, as described in greater detail below. In some implementations, the base support **302**, arm support **304**, and tilt member **324** can be considered to be a support linkage having display unit **306** coupled at the distal end of the support linkage. For example, the motor(s) can be controlled by control signals from a control circuit (e.g., control system) to move display unit **306** about tilt axis **326** to a particular orientation in the tilt degree of freedom **327**.

(67) In some implementations, display unit **306** is rotationally coupled to the tilt member **324** and may rotate about a yaw axis (e.g., lateral rotation axis) **330**. For example, this can be lateral or left-right rotation from the point of view of a user viewing images of display unit **306** via viewports **323**. In the example of FIGS. **3-5**, display unit **306** is coupled to the tilt member by a rotary mechanism which, in some implementations, can include a track mechanism. For example, in some implementations, the track mechanism includes a curved track **328**, and curved track **328** is coupled to display unit **306**. Track **328** slidably engages a groove member coupled to tilt member **324**, allowing display unit **306** to rotate about yaw axis **330** by moving curved track **328** through a groove of the groove member. In some implementations, a curved track is coupled to tilt member **324** and a groove member is coupled to display unit **306**, and the groove member engages and slides along the length of the curved track to allow the rotary movement of display unit **306** about yaw axis **330**. In some implementations, the groove member can be about as long as the width of tilt member **324** and/or include at least a portion of a loop through which the curved track **328** slides.

(68) In some implementations, curved track **328** is a curved rail that slidably engages a groove member as described. In some implementations, curved track **328** can be a curved cam follower that engages a cam roller. For example, the cam roller can be rotatably coupled to display unit **306** and, in various implementations, have an axis of rotation that is orthogonal to the yaw axis **330**, or parallel to the yaw axis **330**. For example, the cam roller can be cylindrical and can roll along the curved surface of the cam follower that is rigidly coupled to tilt member **324**. For example, the cam roller can be held against the cam follower by walls or ridges of the cam follower. In some implementations, the cam follower can be rigidly coupled to the display unit **306**, and the cam roller can be rotatably coupled to tilt member **324**.

(69) The curvature (e.g., radius) of the curved track **328** and/or groove member is selected to provide the yaw axis **330** at a particular distance from a user-facing side of display unit **306** and/or from tilt axis **326**. For example, this can be a particular horizontal distance that is parallel to the degree of freedom **322** of the arm portion **320** in some implementations. For example, the yaw axis **330** can be provided at a particular distance from display unit **306** such that it approximately intersects a defined (e.g., virtual or software-defined) neck pivot axis corresponding to a pivot axis in a user's neck, as described below. The defined neck pivot axis can be used as a reference for motion of display unit **306** in some implementations. In the described implementation, the angle between yaw axis **330** and a vertical axis (e.g., parallel to degree of freedom **316**) varies based on the orientation of the tilt member **324** about tilt axis **326**. The yaw motion of display unit **306** about yaw axis **330** can be driven by one or more actuators, e.g., motors, using a drive mechanism such as a gear mechanism, capstan drive mechanism, etc. An example of a capstan drive mechanism is described in greater detail below with respect to FIG. **22**. For example, a drive mechanism can include a capstan drum **329** that is coupled to a capstan pulley driven by a motor.
(70) In some implementations, other couplings or bearings can be used to provide rotational motion of display unit **306** about yaw axis **330** with respect to the tilt member **324** and arm support **304**. For example, a rotary joint similar to the rotary coupling between tilt member **324** and arm support **304** can be used to couple the display unit **306** to tilt member **324**. In further examples, a vertically-aligned track mechanism that provides rotation about a horizontal axis, e.g., similar to members **2324** and **2330** of implementations of FIGS. **23-25**, can provide a tilt degree of freedom to display unit **306** and a rotary coupling can provide a yaw degree of freedom to display unit **306**.
(71) Display system **300** thus provides display unit **306** with vertical linear degree of freedom **316**, horizontal linear degree of freedom **322**, rotational (tilt) degree of freedom **327**, and rotational yaw degree of freedom **331**. For example, the vertical and horizontal degrees of freedom allow the display unit **306** to be moved to any position within an allowed workspace (e.g., in a vertical plane), and the tilt degree of freedom allows the display unit to be moved to a particular orientation within its range of motion (e.g., within the vertical plane or a parallel vertical plane).
(72) A combination of coordinated movement of components of display system **300** in at least two of these degrees of freedom allow display unit **306** to be positioned at various positions and orientations in its workspace, e.g., translated and/or rotated around a user, to facilitate a custom viewing experience for the user using the display unit. The motion of display unit **306** in the tilt, horizontal, and/or vertical degrees of freedom allows display unit **306** to stay close to the user's head and eyes during user head motion, and/or maintain a physical connection between the user's head (e.g., forehead) and display unit **306**.
(73) For example, display unit **306** is positionable (e.g., translatable and/or rotatable) in its workspace such that eyes of the user align with the viewports of the display unit. In addition, display unit **306** can be rotated in physical space about a defined eye pivot axis corresponding to (e.g., coincident with), for example, an eye axis through both of a user's eyes to allow a desired vertical (e.g., up-down) eye viewing angle for the user. The display unit can also be moved about yaw axis **330** to allow a desired yaw (e.g., left-right) viewing angle or orientation for the user. These rotations allow the display unit **306** to be oriented comfortably for the user to view images through the viewports.
(74) The degrees of freedom of the display system also or alternatively allow the display system **300** to provide motion of display unit **306** in physical space about a different defined pivot axis that can be positioned in any of various locations in the workspace of the display unit **306** (examples described in greater detail below). For example, the system **300** can provide motion of display unit **306** in physical space that corresponds to motion of a user's head when operating the display system **300**. This motion can include rotation about a defined neck pivot axis that approximately corresponds to a neck axis of the user's head at the user's neck. This rotation allows the display unit **306** to be moved in accordance with the user's head that is directing movement of the display unit

306, e.g., using a head input device **342**. Some examples of such motion of a display unit about a neck pivot axis (FIGS. **8-11**) and an eye pivot axis (FIGS. **12-14**) are described below.
(75) In another example, the motion of display unit **306** can include rotation about a defined forehead pivot axis that approximately corresponds to a forehead axis extending through the user's head at the user's forehead when the display unit **306** is oriented, as shown, in a centered yaw rotary position about yaw axis **330**. In some implementations, the forehead pivot axis corresponds to a forehead axis extending through a portion of an input device of the display unit **306** (e.g., head input device **342**), where the portion is at or near a point of contact between the user's forehead and the input device. In some implementations, the defined forehead pivot axis can be oriented parallel to tilt axis **326**, e.g., orthogonal to the linear degrees of freedom **316** and **322**. The forehead pivot axis can alternately be positioned to correspond to a different location or portion of the user's forehead or display unit.
(76) In another example, the motion of display unit **306** can include rotation about a defined hand input device pivot axis that approximately corresponds to an axis that extends through one or more hand input devices of display unit **306** (examples described below). For example, this axis can extend through portions (e.g., the centers of grips) of both hand input devices **340***a* and **340***b* that are positioned on opposing (left and right) sides of display unit **306**. In some implementations, the defined hand input device pivot axis can be oriented parallel to tilt axis **326**, e.g., orthogonal to the linear degrees of freedom **316** and **322**, similarly to the neck, eye, and forehead pivot axes described above. The hand input device pivot axis can alternately be positioned to correspond to a different location or portion of the display unit, e.g., a location of a different hand input device. For example, the hand input device pivot axis provides an axis of rotation about which the display unit **306** can be commanded to be rotated by manipulation of the hand input devices **340***a* and **340***b* by the user, and/or by user manipulation of other user input devices such as head input device **342**, control input devices **210** and **212**, etc.
(77) In another example, the motion of display unit **306** in its workspace can include linear motion, e.g., based on linear translation in the vertical linear degree of freedom **316** and horizontal linear degree of freedom **322** and without rotational motion in the tilt and/or yaw degrees of freedom **327** and/or **331**. In another example, the motion can include both linear motion and rotational motion. For example, display unit **306** can be moved linearly and then rotationally in its workspace, and/or vice-versa.
(78) During rotation of the display unit **306** about a defined pivot axis, the movement of base support **302** in vertical degree of freedom **316** and/or the movement of arm support **304** in horizontal degree of freedom **322** can have a change in direction without a change in direction in rotation of the display unit about the defined pivot axis. For example, arm support **304** can reverse direction as the display unit is rotated from fully up to fully down positions. Thus, it is possible in some implementations for the base support or arm support to end up in the same position while the display unit has rotated from a first orientation to a second orientation about the defined pivot axis, where the support moves back and forth while the display unit transitions between those two orientations.
(79) Display unit **306** can include input devices that allow a user to provide input to manipulate the orientation and/or position of the display unit **306** in space, and/or to manipulate other functions or components of the display system **300** and/or a larger system (e.g., teleoperated system).
(80) For example, hand input devices **340** can be provided on display unit **306** similarly as described for FIG. **2** (shown as **206** in FIG. **2**). For example, a hand input device **340***a* is positioned on a left side of the display unit **306** and a hand input device **340***b* on a right side of the display unit **306**, or can be positioned on any surface of the display unit **306**. The hand input devices **340***a* and **340***b* can be any of various types of user input devices (e.g., buttons, touchpads, force sensors, joysticks, knobs, trackballs, keyboards, etc.) manipulated by a user's hands to provide user hand input to the hand input device **340**. The hand input device **340** outputs a control signal to the

display system **300** based on the user hand input. The control signal can be used to cause display unit **306** to change its orientation and/or position in space. For example, the control system controls actuators of the display system **300** to move the base portion **314** in linear degree of freedom **316**, arm portion **320** in linear degree of freedom **322**, tilt member **324** in rotary degree of freedom **327**, and/or display unit **306** in rotary degree of freedom **331**, to cause the display unit **306** to be moved corresponding to the sensed user hand input. Sensed user hand input can also be used to control other functions of the display system **300** and/or functions of a larger system (e.g., teleoperated system **100** of FIG. **1**).

(81) Display unit **306** can include a head input device **342** similarly as described for FIG. **2** (shown as **208** in FIG. **2**). In this example, head input device **342** is positioned on a surface of the display unit **306** that is facing the user's head during operation of the display unit **306**. For example, head input device **342** can include one or more sensors that sense user head input that is received as commands to cause the display unit **306** to change its position and/or orientation in space. In some implementations, sensing the user head input can include sensing a presence or contact by a user's head or portion of the head (e.g., forehead) with the head input device **342**. In some examples, the one or more sensors can include any of a variety of types of sensors, e.g., resistance sensors, capacitive sensors, force sensors, optical sensors, etc.

(82) The orientation and/or position of the display unit **306** can be changed by the display system **300** based on the user head input to head input device **342**. For example, sensed user input is provided to a control system, which controls actuators of the display system **300** to move the base portion **314** in linear degree of freedom **316**, arm portion **320** in linear degree of freedom **322**, tilt member **324** in rotary degree of freedom **327**, and/or display unit **306** in rotary degree of freedom **331**, to cause the display unit **306** to be moved as commanded by (e.g., in accordance with) the sensed user head input. For example, sensed head input can be used to rotate display unit **306** about a defined pivot axis, e.g., a neck pivot axis or other pivot axis as described herein. In some implementations, sensed head input can be additionally or alternatively used to rotate display unit **306** about yaw axis **330**. Sensed user head input can also be used to control other functions of the display system **300** and/or of a larger system (e.g., teleoperated system **100** of FIG. **1**). Thus, in some implementations, the user can move his or her head to provide input to input device to control the display unit **306** to be moved by the display system in accordance with the motion of the head, thus allowing the display unit to follow motions of the user's head and changes in viewing angle, e.g., without having to attach the display unit to the head of the user and without requiring the user to physically move the display unit. In some implementations, moving the user's head away from display unit **306** (e.g., to “pull” the display unit **306** toward the user who is moving back) can be sensed as movement in a direction away from the head input device **342** or away from display unit **306** (e.g., sensed via optical sensor or other type of sensor), or sensed as a reduction of force on the head input device **342**, which causes the display system to move the display unit via control of actuators to follow the user's head.

(83) In some implementations, display unit **306** can additionally or alternatively be moved in one or more of its described degrees of freedom in response to receiving user input from other input devices of display system **300** or of other connected systems. For example, additional hand input devices can be provided such as control input devices **210** and **212** of FIG. **2** or other control devices. In some examples, input devices can be coupled to a support surface that is coupled to base support **302**, or coupled to ergonomic support **214**, etc. In some implementations, user input received at foot controls **220** or other types of input devices can be used to move the display unit **306** in one or more of its degrees of freedom.

(84) In some implementations, images displayed by the display unit **306**, and/or other controlled devices, are changed and manipulated based on the sensed motion of the display unit **306**, e.g., in the workspace of the display unit, about a defined pivot axis, etc. For example, motion of the display unit up and down about a defined neck pivot axis can modify display images or view in a

corresponding direction.
(85) In some implementations of a display system, display unit **306** is rotatable about yaw axis **330** in degree of freedom **331** and one or more of the other degrees of freedom **316**, **322**, and/or **327** are omitted from the display system **300**. For example, display unit **306** can be rotated about yaw axis **330** (e.g., by actuator(s) and/or manually by a user) and the display unit **306** can be manually positioned higher and/or lower (e.g., by actuator(s) and/or manually by a user), e.g., using base support **302** or other mechanism, where horizontal degree of freedom **322** and/or tilt degree of freedom **327** are omitted.
(86) In some example implementations, a control unit comprises a first support, a second support coupled to the first support, and a display unit rotatably coupled to the second support. The display unit and the second support are linearly translatable along a first axis in a first degree of freedom with respect to the first support, the display unit is linearly translatable along a second axis in a second degree of freedom with respect to the first support and the second support, and the display unit is rotatable about a third axis with respect to the second support in a third degree of freedom. In further examples, a tilt member couples the display unit to the second support, the tilt member being rotatable about a fourth axis with respect to the tilt member.
(87) FIG. **6** is a side view of another example of a portion of display system **600**. Display system **600** includes similar components to the display system **300** described above.
(88) In the example of FIG. **6**, second arm portion **620** of arm support **604** includes a distal portion **623** which extends vertically lower than the distal portion of the second arm portion **320** of display system **300** of FIGS. **3-5**. Tilt axis **626** of tilt member **624** is located at the end of the distal portion **622**. In some implementations, the extended distal portion **622** may allow tilt axis **626** to be positioned lower than the tilt axis **326** of display system **300** and closer to a defined (virtual or software-defined) pivot axis around which the display unit is to be rotated. For example, the defined pivot axis can be any of the example defined pivot axes herein, e.g., an eye pivot axis or neck pivot axis of a user **650** operating the display unit **606**, examples of which are shown with respect to FIGS. **8-11**.
(89) Providing tilt axis **626** closer to a defined pivot axis may require less movement of the components of the display system **600** to rotate the display unit **606** about that defined pivot axis, as compared to a tilt axis located further from the defined pivot axis. For example, less movement may be required for second arm portion **620**, second base portion **614**, and/or tilt member **624**, allowing a reduction in cost in some implementations due to reduced component size, motor size, etc.
(90) FIG. **7** is a side view of yet another example of a portion of display system **700**. Display system **700** includes similar components to the display systems **300** and **600** described above.
(91) In the example of FIG. **7**, second arm portion **720** of arm support **704** includes a distal portion **723** which extends vertically lower than the distal portion of the second arm portion **320** of display system **300** of FIGS. **3-5**. Distal portion **722** includes an end that extends both vertically down and toward the base support **702**, e.g., away from a user location that is occupied by a user **750** when operating the display system. Tilt axis **726** is located at the end of the distal portion **722**. This position of tilt axis **726** is moved further downward, and further away from the user position, as compared to tilt axis **326** of display system **300** and tilt axis **626** of display system **600**.
(92) The position of tilt axis **726** may have advantages in some implementations. For example, the end of the distal portion **723** can be further from the user **750**, providing the user with a more open-feeling area in which to operate the display unit **706**. Furthermore, this position can allow a reduction in the maximum gravity load on the joint of the tilt axis **726**, and can also reduce the gravity load of display unit **706** in particular rotary positions about tilt axis **726**, e.g., at higher tilt positions (e.g., view angles oriented horizontally and downward) where display unit **706** may be often situated during operation. For example, the position of tilt axis **726** may be closer to a position directly above a center of gravity of the component that includes display unit **706** and the

track mechanism coupling display unit **706** to the tilt member **724**. The position of tilt axis **726** also changes the amount of movement required by the second arm portion **720**, second base portion **714**, and/or tilt member **724** as compared to the tilt axis positions of display systems **300** and **600**. For example, the position of tilt axis **726** may require less horizontal movement in degree of freedom **722** and more vertical movement in degree of freedom **716** than in display systems **300** and **600**, which may be advantageous in some implementations, e.g., for enabling greater stiffness of the display system, operating within component or environment limitations, etc.

(93) In addition, as shown in FIG. **7**, tilt member **724** can be reduced in length compared to the lengths of tilt members **324** and **624** described in display systems **300** and **600**. In addition, the distance can be reduced between the tilt axis **724** and the center of gravity of the component that includes display unit **706** and the track mechanism coupling display unit **706** to the tilt member **724**. These smaller distances can reduce the joint side inertia of tilt motion of display unit **706** about tilt axis **726**. In some implementations, the decrease in length of the tilt member **724** can also improve the stiffness and reduce the weight of the overall support structure.

(94) In some implementations, the above features of display system **700** can result in lower torque output requirements, lower average current use, and lower power dissipation by a motor or other actuator used to drive rotation of the display unit **706** about the tilt axis **726**, and may allow additional operating headroom for the actuator and/or drive gearing mechanism (allowing the actuator to output forces greater than typically required for operation of the display system **700**).

(95) FIGS. **8-11** are side views a portion of a display system **800** showing rotation of a display unit about an example defined neck pivot axis, according to some implementations. Display system **800** includes similar components to the display systems **300** and **600** described above.

(96) In FIG. **8**, display system **800** is shown having a display unit in a first pivot orientation. Display system **800** includes a base support **802**, an arm support **804**, and a display unit **806** similarly as described above. In this example, base support **802** includes a second base portion **814** that is linearly translatable through an interior of a first base portion **812** in a linear degree of freedom **816**. Arm support **804** includes a first arm portion **818** rigidly coupled to second base portion **814**, and a second arm portion **820** that is linearly translatable through an interior of first arm portion **818** in a linear degree of freedom **822**.

(97) Display unit **806** is coupled to the second arm portion **820** by a tilt member **824** similarly as described in examples above. Tilt member **824** is rotationally coupled at a first end to the second arm portion **820** and rotates about axis **826**. Display unit **806** can be rotationally coupled to a second end of the tilt member **824** similarly as described in examples above, and rotates about axis **826** in accordance with tilt member **824**.

(98) In FIG. **8**, display unit **806** is oriented in a first pivot orientation about an example defined neck pivot axis **840** that can be approximately horizontal and can extend approximately parallel to tilt axis **826**, e.g., orthogonal to a plane defined by the degrees of freedom **816** and **822**. In the example implementations shown, neck pivot axis **840** is located below the tilt axis **826**. In some implementations, the defined neck pivot axis **840** is located further away from the base support **802** than the display unit **806**.

(99) The defined neck pivot axis **840** (and any of the defined neck pivot axes described herein) is oriented at a location that is aligned (e.g., approximately aligned) with a pivot axis of a neck of a typical user such as user **850** when the display unit **806** is oriented, as shown, in a centered yaw rotary orientation about yaw axis **830**. For example, the defined neck pivot axis can be a horizontal axis that is positioned to intersect a location at which a neck of a typical user operating the display unit approximately pivots. This location may be different in different users (e.g., users of different heights, different sizes of necks, etc.), and/or users may have different preferences as to where the defined neck pivot axis is located, such that in some implementations, a location determined (e.g., averaged) from preferred pivot locations of multiple users can be used for the defined neck pivot axis. In some example implementations, the defined neck pivot axis can be located at a location

that is approximately aligned with a particular bone or any particular cervical vertebra of a neck of a user operating the display unit, e.g., an atlas bone or axis bone of a neck. In further examples, the defined neck pivot axis can be defined to correspond to other locations of a user's neck.

(100) The neck pivot axis **840** is a virtual axis that is defined based on system parameters. The movement of the base support **802**, arm support **804**, and tilting member **824** can provide the rotation about neck pivot axis **840**. Neck pivot axis **840** can be at different positions than shown in FIG. **8**, as defined by the movement of these components in creating rotation about the desired neck pivot axis.

(101) Some implementations may allow the location of the defined neck pivot axis to be changed, e.g., by user input, to accommodate the specific preferences or physiology of particular users. In some examples, the defined neck pivot axis can be changed to a different location at which the defined neck pivot axis is parallel to tilt axis **826** (such as a different location in which the axis is orthogonal to a vertical plane defined by the degrees of freedom **816** and **822**), e.g., a new location that is up, down, forward, and/or back from a previous location in the point of view of a user. In some implementations, a stored profile or settings can be stored in association with a particular user and applied for use of the display system by that user. The stored profile includes preferred locations in space for the defined neck pivot axis and/or defined eye pivot axis (described below), any of which can be loaded for system operation. Some implementations can guide a user in determining a defined neck pivot axis customized for that user. For example, the system can output instructions to the user to bend their neck, straighten their neck, etc., and sense the rotation and trajectory of the user's head in relation to the user's body based on sensor data (e.g., images captured by an image capture device). This data is used by the display system (or control system) to determine a location of a suggested defined neck pivot axis for that user, a description of which can be output by an output device such as the display device of the display unit **806**.

(102) In the first pivot orientation shown, display unit **806** is in a horizontal view orientation about the defined pivot axis **840**. A view orientation of display unit **806** with respect to neck pivot axis **840** is indicated by a line **842** that, in FIG. **8**, extends horizontally through the neck pivot axis **840**. In this implementation, the horizontal view orientation corresponds to tilt member **824** in a horizontal orientation parallel to arm support **804**. Rotation of display unit **806** about the neck pivot axis **840** is achieved by moving portion **814** of the base support **802**, portion **820** of the arm support **804**, and/or tilt member **824** to appropriate positions and/or orientations.

(103) FIG. **8** also shows a defined eye pivot axis **860**. The defined eye pivot axis **860** is positioned at a location that corresponds to (e.g., is coincident with) an eye axis that intersects the eyes of a typical user such as user **850** that is looking through the viewports of the display unit **806** when the display unit **806** is oriented, as shown, in a centered yaw rotary orientation about yaw axis **830**. In some implementations, the defined eye pivot axis extends parallel to tilt axis **826**, e.g., orthogonal to a plane defined by the degrees of freedom **816** and **822**. This defined eye pivot axis can be orthogonal to the view orientation of the display unit **806** when the display unit **806** is oriented, as shown, in a centered yaw rotary orientation about axis **830**. A view orientation of display unit **806** with respect to eye pivot axis **860** is indicated by a sight line **862** that, in FIG. **8**, extends horizontally through eye pivot axis **860** and indicates a reference view angle for user **850** of the display unit. In some implementations, as shown, the tilt axis **826** is located above the sight line **862**. Movement of the display system **800** based on the eye pivot axis **860** is described in greater detail below with respect to FIGS. **12-14**. Movement of the display system **800** about other defined pivot axes of the display unit **306**, e.g., forehead pivot axis or hand input device axis, can be similarly implemented to the movement about the neck and/or eye pivot axes.

(104) In FIG. **9**, display unit **806** is oriented at a second pivot orientation rotated about the neck pivot axis **840** to allow an “upward” view angle of the user **850**. An angle A**1** is an example angle of rotation of the display unit **806** about the neck pivot axis **840**, where angle A**1** is the angle between the horizontal view orientation **842** of the display unit **806** and a neck-axis view

orientation **942** of display unit **806**. Display unit **806** has been rotated about the neck pivot axis **840** relative to the first pivot orientation shown in FIG. **8** by moving the base support **802**, arm support **804**, and tilt member **824** to positions and/or orientations as shown. For example, relative to the first pivot orientation shown in FIG. **8**, the display unit **806** is moved to the second pivot orientation by rotating tilt member **824** upward to the angle shown (counterclockwise about axis **826** in the viewpoint of FIG. **9**), linearly translating second arm portion **820** away from base support **802** and toward the user **850** to the position shown, and linearly translating second base portion **814** upward (e.g., away from ground) to the position shown. The tilt axis **826** is independent of the neck pivot axis **840**, since rotation about the neck pivot axis **840** uses rotation about the tilt axis **826** as well as linear motion (translation) in the degrees of freedom **822** and **816**.
(105) In some implementations, the view angle shown in FIG. **9** is the upward (e.g., counterclockwise as shown in FIG. **9**) rotational limit of the display unit **806**. In other implementations, display unit **806** can be rotated upward (e.g., counterclockwise) by greater amounts than shown in FIG. **9** before reaching a limit to rotation in this direction.
(106) In FIG. **10**, display unit **806** is oriented at a third pivot orientation rotated about the defined neck pivot axis **840** to allow a “downward” view angle of the user **850**. An angle A**2** is an example angle of rotation of the display unit **806** about the neck pivot axis **840**, where angle A**2** is the angle between the horizontal view orientation **842** of the display unit **806** and a neck-axis view orientation **1042** of the display unit. Display unit **806** has been rotated about the neck pivot axis **840** by moving the base support **802**, arm support **804**, and tilt member **824** to positions and/or orientations as shown. For example, relative to the first pivot orientation shown in FIG. **8**, the display unit **806** is moved to the third pivot orientation by rotating tilt member **824** downward to the angle shown (clockwise about axis **826** in the viewpoint of FIG. **10**), linearly translating second arm portion **820** in degree of freedom **822** toward the base support **802** and away from the user **850** to the position shown, and linearly translating second base portion **814** downward (e.g., toward ground) to the position shown.
(107) In FIG. **11**, display unit **806** is oriented at a fourth pivot orientation rotated about the defined neck pivot axis **840** to allow a further downward view angle of the user **850**. An angle A**3** is an example angle of rotation of the display unit **806** about the neck pivot axis **840**, where angle A**3** is the angle between the horizontal view orientation **842** of the display unit **806** and a neck-axis view orientation **1142** of the display unit. Display unit **806** has been rotated about the neck pivot axis **840** by moving the base support **802**, arm support **804**, and tilt member **824** to positions and/or orientations as shown. For example, relative to the third pivot orientation shown in FIG. **10**, the display unit **806** is moved to the fourth pivot orientation by rotating tilt member **824** further downward to the angle shown (clockwise about axis **826** in the viewpoint of FIG. **11**), linearly translating second arm portion **820** in degree of freedom **822** toward the base support **802** and away from the user **850** to the position shown, and linearly translating second base portion **814** downward (e.g., toward ground) to the position shown.
(108) In the example implementations shown in FIGS. **8-11**, the distance between defined neck pivot axis **840** and tilt axis **826** is fixed during rotation of the display unit about the defined neck pivot axis **840**.
(109) FIGS. **12-14** are side views of a portion of display system **800** showing rotation of a display unit about a defined eye pivot axis, according to some implementations. Display system **800** includes similar components to the display systems **300**, **600**, and **700** described above.
(110) In FIG. **12**, display unit **806** is oriented at a fifth pivot orientation, which is rotated about the defined eye pivot axis **860** to allow an upward view angle of the user **850**. An angle A**4** is an example angle of rotation of the display unit **806** about the eye pivot axis **860**, where angle A**4** is the angle between the horizontal eye-axis view orientation **862** of the display unit **806** (as shown in FIG. **8**) and an eye-axis view orientation **1262** of the display unit. Display unit **806** has been rotated about the eye pivot axis **860** relative to the first pivot orientation shown in FIG. **8** by moving the

base support **802**, arm support **804**, and tilt member **824** to positions and/or orientations as shown. For example, relative to the first pivot orientation shown in FIG. **8**, the display unit **806** is moved to the fifth pivot orientation by rotating tilt member **824** upward to the angle shown (counterclockwise about axis **826** in the viewpoint of FIG. **12**), linearly translating second arm portion **804** away from base support **802** and toward the user **850** to the position shown, and linearly translating second base portion **814** upward (e.g., away from ground) to the position shown. The tilt axis **826** is independent of the eye pivot axis **860**, since rotation of the display unit **806** about the eye pivot axis **860** uses rotation about the tilt axis **826** as well as linear motion (translation) in the degrees of freedom **822** and **816**.

(111) In FIG. **13**, display unit **806** is oriented at a sixth pivot orientation, which is rotated about the defined eye pivot axis **860** to allow a downward view angle of the user **850**. An angle A**5** is an example angle of rotation of the display unit **806** about the eye pivot axis **860**, where angle A**5** is the angle between the horizontal eye-axis view orientation **862** of the display unit **806** and an eye-axis view orientation **1362** of the display unit. Display unit **806** has been rotated about the eye pivot axis **860** by moving the base support **802**, arm support **804**, and tilt member **824** to positions and/or orientations as shown. For example, relative to the first pivot orientation shown in FIG. **8**, the display unit **806** is moved to the sixth pivot orientation by rotating tilt member **824** downward to the angle shown (clockwise about axis **826** in the viewpoint of FIG. **13**), linearly translating second arm portion **804** in degree of freedom **822** toward the base support **802** and away from the user **850** by the distance shown, and linearly translating second base portion **814** downward (e.g., toward ground) by the distance shown.

(112) In FIG. **14**, display unit **806** is oriented at a seventh pivot orientation, which is rotated about the defined eye pivot axis **860** to allow a further downward view angle of the user **850**. An angle A**6** is an example angle of rotation of the display unit **806** about the eye pivot axis **860**, where angle A**6** is the angle between the horizontal eye-axis view orientation **862** of the display unit **806** and an eye-axis view orientation **1462** of the display unit. Display unit **806** has been rotated about the eye pivot axis **860** by moving the base support **802**, arm support **804**, and tilt member **824** to positions and/or orientations as shown. For example, relative to the first pivot orientation shown in FIG. **8**, the display unit **806** is moved to the seventh pivot orientation by rotating tilt member **824** downward to the angle shown (clockwise about axis **826** in the viewpoint of FIG. **14**), linearly translating second arm portion **804** in degree of freedom **822** toward the base support **802** and away from the user **850** to the position shown, and linearly translating second base portion **814** downward (e.g., toward ground) to the position shown.

(113) FIGS. **15** and **16** are perspective and side views, respectively, of an example base support **1500** for a display system, according to some implementations. For example, base support **1500** can be used as base support **302** of display system **300** (FIGS. **3-5**), as base support **602** of display system **600** (FIG. **6**), or as base support **702** of display system **700** (FIG. **7**).

(114) Base support **1500** includes a first base portion **1502** and a second base portion **1504** which are telescoping. First base portion **1502** is an elongated member that is coupled to ground, e.g., a floor or other support surface, and includes a hollow interior. Second base portion **1504** is an elongated member that is positioned in the interior of first base portion **1502** such that it is slidably engaged with the first base portion **1502**. In some implementations, second base portion **1504** can include lengthwise (e.g., vertical) grooves that are engaged by mating portions of first base portion **1502**, or vice-versa, to provide stability of the sliding engagement of the first and second base portions.

(115) A drive mechanism applies force to second base portion **1504** to move this portion. The drive mechanism includes one or more actuators to linearly translate the second base portion **1504** within first base portion **1502** in the degree of freedom **1506**. In this example, the drive mechanism includes a ballscrew transmission to transmit force from an actuator to the second base portion **1504**. The actuator can be a rotary motor **1508** that is rigidly coupled to one end of the first base

portion **1502**, e.g., the end that is coupled to a grounded support. Motor **1508** has a rotating shaft that is coupled to a first end of a ballscrew **1510** that extends through the interior portion of the first base portion **1502** and through the interior portion of the second base portion **1504**. Ballscrew **1510** is a threaded member that engages a threaded aperture of ballscrew nut **1512**. Ballscrew nut **1512** is rigidly coupled to the second base portion **1504**. In some implementations, ballscrew **1510** can be rotatably coupled at its second end to a support bearing **1514**, where support bearing **1514** is rigidly coupled to the second base portion **1504**. For example, support bearing **1514** can align ballscrew **1510** along a central axis through ballscrew nut **1512** parallel to ballscrew **1510**. In some implementations, support bearing **1514** is not used.

(116) In operation, the motor **1508** outputs a force on its rotatable shaft to cause the ballscrew **1510** to rotate. The rotation of the ballscrew **1510** causes the ballscrew nut **1512** and support bearing **1514** to move linearly along the longitudinal axis of the ballscrew **1510**. The movement of ballscrew nut **1512** is constrained by the ballscrew **1510** and by the sliding engagement of first and second base portions **1502** and **1504**. The linear motion of the ballscrew nut **1512** moves the second base portion **1504** linearly in the degree of freedom **1506**. The ballscrew transmission thus converts rotary force from motor **1508** to linear force applied to the second base portion **1504**. In some implementations, motor **1508** can be coupled to a sensor such as a rotary encoder that determines a rotational orientation of the motor shaft and ballscrew **1510**, which can be converted to a linear position of the second base portion **1504** with respect to the first base portion **1502**.

(117) Motor **1508** can be any of a variety of types of actuators similarly as described herein for other implementations. For example, active actuators can be used, e.g., motors (e.g., DC motors), voice coils, or other types of active actuators. Motor **1508** can be controlled using control signals from a control circuit, e.g., of a teleoperated system or other system. One or more sensors can be coupled to one or more of the components of FIG. **15** which are operative to detect the translation and/or position of the second base portion **1504** with respect to the first base portion **1502**. For example, in some implementations, in addition to or instead of rotary encoder coupled to the motor **1508**, a linear sensor can be coupled to the first base portion **1502** and/or second base portion **1504** to sense linear motion of the second base portion **1504** with respect to the first base portion **1502**. The sensors can send signals describing sensed positions, orientations, or motion to one or more control circuits, e.g., a control system of the teleoperated system **100**. In some modes or implementations, the control circuits can provide control signals indicating the sensed positions, orientation, or motion to a manipulator system. The sensors can be any of a variety of types of sensors, e.g., a magnetic sensor (e.g., magnetic incremental linear position sensor, Hall Effect sensor, etc.), optical sensor, encoder, resistance sensor, etc.

(118) In some implementations, an additional drive mechanism can be provided on base support **1500** and coupled to an ergonomic support, e.g., an ergonomic support **214** of FIG. **2**, to provide forces to cause linear up and down motion of the ergonomic support and accommodate particular user preferences for support height. In some examples, the additional drive mechanism can be mounted to the outside of the base support **1500** while the drive mechanism for second base portion **1504** can be provided on the inside of the base support **302** as shown. For example, the additional drive mechanism can use an actuator and ballscrew mechanism similarly as described above, and/or a different drive mechanism.

(119) FIGS. **17** and **18** are perspective and side views, respectively, of an example arm support **1700** for a display system, according to some implementations. For example, arm support **1700** can be used as arm support **304** of display system **300** (FIGS. **3-5**), as arm support **604** of display system **600** (FIG. **6**), or as arm support **704** of display system **700** (FIG. **7**).

(120) Arm support **1700** includes a first arm portion **1702** and a second arm portion **1704** which are telescoping. First arm portion **1702** is an elongated member with a hollow interior and, in some implementations, can be coupled to a base support, e.g., second base portion **314**, **614**, or **714**, or second base portion **1504** of FIG. **15**. Second arm portion **1704** is an elongated member that is

positioned in the interior of first arm portion **1702** such that it is slidably engaged with the first arm portion **1702**. In some implementations, second arm portion **1704** can include lengthwise (e.g., horizontal) grooves that are engaged by mating portions of first arm portion **1702**, or vice-versa, to provide stability of the sliding engagement of the first and second arm portions.

(121) Second arm portion **1704** is driven by one or more actuators to linearly translate within first arm portion **1702** in the degree of freedom **1706**. In this example, a ballscrew transmission is used to transmit force from an actuator to the second arm portion **1704**. The actuator can be a rotary motor **1708** that is rigidly coupled to one end of the first arm portion **1702**, e.g., the end that is coupled to a grounded support. Motor **1708** has a rotating shaft that is coupled to a first end of a ballscrew **1710** that extends through the interior portion of the first arm portion **1702** and through the interior portion of the second base portion **1704**. Ballscrew **1710** is a threaded member that engages a threaded aperture of ballscrew nut **1712**. Ballscrew nut **1712** is rigidly coupled to the second arm portion **1704**. In some implementations, ballscrew **1710** can be rotatably coupled at its second end to a support **1714**, and support **1714** is rigidly coupled to the second arm portion **1704** and aligns ballscrew **1710** along a central axis through the ballscrew nut **1712** parallel to the ballscrew **1710**. In some implementations, support bearing **1714** is not used.

(122) In operation, the motor **1708** outputs a force on its rotatable shaft to cause the ballscrew **1710** to rotate. The rotation of the ballscrew **1710** causes the ballscrew nut **1712** to move linearly along the longitudinal axis of the ballscrew **1710**. The movement of the ballscrew nut **1712** is constrained by the ballscrew **1710** and by the sliding engagement of first and second arm portions **1702** and **1704**. The linear motion of the ballscrew nut **1712** moves the second arm portion **1704** linearly in the degree of freedom **1706**. The ballscrew transmission thus converts rotary force from motor **1708** to linear force applied to the second arm portion **1704**. In some implementations, motor **1708** can be coupled to a sensor such as a rotary encoder that determines a rotational orientation of the motor shaft and ballscrew **1710**, which can be converted to a linear position of the second arm portion **1704** with respect to the first arm portion **1702**.

(123) Motor **1708** can be any of a variety of types of actuators similarly as described herein for other implementations, e.g., active actuators such as motors (e.g., DC motors), voice coils, or other types of active actuators. Motor **1708** can be controlled using control signals from a control circuit, e.g., of a teleoperated system or other system. One or more sensors can be coupled to one or more of the components of FIG. **17** which are operative to detect the translation and/or position of the second arm portion **1704** with respect to the first arm portion **1702**. For example, in some implementations, in addition to or instead of rotary encoder coupled to the motor **1708**, a linear sensor can be coupled to the first arm portion **1702** and/or second arm portion **1704** to sense linear motion of the second arm portion **1704** with respect to the first arm portion **1702**. The sensors can send signals describing sensed positions, orientations, or motion to one or more control circuits, e.g., a control circuit of the teleoperated system **100**. In some modes or implementations, the control circuits can provide control signals indicating the sensed positions, orientations, or motion to a manipulator system. The sensors can be any of a variety of types of sensors, e.g., a magnetic sensor (e.g., magnetic incremental linear position sensor, Hall Effect sensor, etc.), optical sensor, encoder, resistance sensor, etc.

(124) FIGS. **19**, **20**, and **21** are side, front, and perspective views, respectively, of an example tilt mechanism **1900**, according to some implementations.

(125) Tilt mechanism **1900** includes a tilt member **1902** and a tilt drive mechanism **1904**. For example, tilt member **1902** can be used as tilt member **324** of the display system **300** (FIGS. **3-5**), as tilt member **624** of the display system **600** (FIG. **6**), or as tilt member **724** of the display system **700** (FIG. **7**). For example, a display unit, such as display unit **306** (FIG. **3**), display unit **606** (FIG. **6**), or display unit **706** (FIG. **7**), can be coupled to the tilt member **1902** similarly as described above. For example, a display unit can be coupled at a portion **1905** of the tilt member **1902**, an example of which is described with reference to FIG. **22**.

(126) Tilt member **1902** is rotatably coupled to a support **1906**. In some examples, support **1906** can be a portion (e.g., distal portion) of, or coupled to, an arm support such as second arm portion **320**, **620**, **720**, or **1704** described above. Tilt member **1902** can rotate about a tilt axis **1908**, which, for example, can be the tilt axis **326**, **626**, or **726** described above.
(127) Tilt drive mechanism **1904** can include one or more actuators to drive the rotation of the tilt member **1902** to a particular orientation about tilt axis **1908**. In this example, a motor **1910** includes a driven shaft **1911** that is rigidly coupled to an extended portion **1912** and/or **1913** of the tilt member **1902**. Motor **1910** has a housing rigidly coupled to the support **1906**. When the shaft **1911** is rotated by the motor **1910**, the tilt member **1902** is rotated about tilt axis **1908**. In some implementations, the driven shaft **1911** of the motor **1910** can be coupled to both first and second extended portions **1912** and **1913** of the tilt member **1902**, or the driven shaft can be coupled to one of the first and second extended portions **1912** or **1913**. In some implementations, the tilt member **1902** includes a single extended portion **1912** or **1913**.
(128) Motor **1910** can be any of a variety of types of actuators, e.g., active actuators such as motors (e.g., DC motors), voice coils, or other types of active actuators. Motor **1910** can be controlled using control signals from a control circuit, e.g., of a teleoperated system or other system.
(129) In some implementations, motor **1910** is coupled to a drive transmission **1914** that can provide gearing of the output rotation of the motor shaft to increase or decrease the amount of rotation of the connected tilt member **1902**, instead of tilt member **1902** being directly rotated the same amount as the motor shaft **1911**. Drive transmission **1914** includes a shaft **1915** that is rigidly coupled to the extended member **1912** of the tilt member **1902**. For example, drive transmission **1914** can provide gearing such that a full rotation of the drive shaft **1911** provides a partial rotation of shaft **1915**, causing a partial rotation of the tilt member **1902**. In some examples, drive transmission **1914** is a harmonic drive mechanism. In some implementations, a different drive transmission can be used, e.g., a capstan drive mechanism, mechanical gears, etc.
(130) One or more sensors can be coupled to one or more of the components of the tilt mechanism **1902** which are operative to detect the rotation of the tilt member **1902** about axis **1908**. For example, in some implementations, a sensor **1916** (e.g., rotary encoder) is coupled to shaft **1911** of motor **1910** and determines a rotational orientation of shaft **1911** so that a rotary orientation of the tilt member **1902** about axis **1908** can be determined (e.g., taking into account rotation reduction provided by drive transmission **1914**). In some implementations, a sensor can be coupled to other components of the tilt mechanism **1900** to sense the rotary orientation of the tilt member **1902** about axis **1908**. The sensor(s) can send signals describing sensed positions, orientations, or motion to one or more control circuits, e.g., a control system of the teleoperated system **100**. In some modes or implementations, the control circuits can provide control signals indicating the sensed positions, orientations, or motion to a manipulator system. The sensors can be any of a variety of types of sensors, e.g., a rotary encoder, a magnetic sensor (e.g., magnetic incremental linear position sensor, Hall Effect sensor, etc.), optical sensor, encoder, resistance sensor, etc.
(131) FIG. **22** is a perspective view of an example display unit mechanism **2200**, according to some implementations. Display unit mechanism **2200** couples a display unit **2202** to a tilt member **2204**. For example, display unit **2202** can be display unit **306**, **606**, or **706** of FIG. **3-5**, **6**, or **7**, respectively. In another example, tilt member **2204** can be tilt member **324**, **624**, or **724** of FIG. **3**, **6**, or **7**, or tilt member **1902** of FIGS. **19-21**.
(132) Display unit mechanism **2200** includes a rotary mechanism rotationally coupling the display unit **2202** to the tilt member **2204** and allowing the display unit **2202** to pivot about an axis **2206** with respect to the tilt member **2204**. In this example, axis **2206** can be perpendicular to the axis **2208** about which the tilt member **2204** rotates. For example, axis **2206** can be axis **330**, **630**, or **730** described above in FIG. **3**, **6**, or **7**, respectively, and axis **2208** can be axis **1908** of FIGS. **19-21**, or tilt axis **326**, **626**, or **726** in FIG. **3**, **6**, or **7**, respectively.
(133) In this example, the rotary mechanism is a track mechanism that includes a curved track

bearing including a curved track **2210** and a groove member **2212**. The curved track **2210** is rigidly coupled to the display unit **2202** and is slidably engaged with a groove or aperture in groove member **2212** that is coupled to or a part of the tilt member **2204**. The curvature of curved track **2210** allows the display unit **2202** to pivot about axis **2206** by sliding the curved track **2210** within the groove of groove member **2212** along the length of the curved track **2210**. In some other implementations, a curved track can be coupled to the tilt member **2204** and a groove member (or a slidably-mating curved rail) can be coupled to display unit **2202**, where the groove member rotates with the display unit **2202** about the axis **2206** along the length of the curved track.

(134) Display unit mechanism **2200** includes a drive mechanism that outputs force to the display unit **2202** and drives the display unit **2202** about axis **2206**. In this example, the drive mechanism includes a capstan drum **2216** which, in some implementations, is rigidly coupled to the curved track **2210** (or in some implementations, can be included as a unitary part of curved track **2210**). The drive mechanism includes a motor **2218** that has a housing rigidly coupled to the tilt member **2204**. Motor **2218** can be any of a variety of types of actuators, e.g., active actuators such as motors (e.g., DC motors), voice coils, or other types of active actuators. Motor **2218** can be controlled using control signals from a control circuit, e.g., of a teleoperated system or other system.

(135) A rotating shaft of the motor **2218** rotates about an axis of rotation **2220** that, for example, can be parallel to the axis **2206** about which the display unit **2202** rotates. The shaft of motor **2218** is rigidly coupled to a capstan pulley **2222** and rotates the capstan pulley **2222** about axis **2220**. In some implementations, capstan pulley **2222** is coupled to the capstan drum **2216** by a cable **2224**. For example, a first end of cable **2224** can be attached to a first end **2226** of the capstan drum **2216**, routed along the side surface of the capstan drum **2216**, wrapped around capstan pulley **2222**, and routed along the surface of the capstan drum **2216** to second end **2228** where the second end of the cable is attached. In some implementations, two cables can be used, e.g., a first cable having a first end attached to drum first end **2226** and a second end attached to capstan pulley **2222**, and a second cable having a first end attached to drum second end **2228** and a second end attached to capstan pulley **2222**, the two cables being routed along the side surface of the capstan drum **2216**.

(136) In operation, motor **2218** is controlled to rotate its shaft about axis **2220**, causing rotation of capstan pulley **2222**. The rotation of capstan pulley **2222** causes movement of cable(s) **2224** via winding and unwinding of the cable(s) **2224** around the capstan pulley **2222**. The movement of the cable(s) **2224** causes display unit **2202** to rotate about axis **2206** as guided by the curved track **2210** in the groove or aperture of groove member **2212**.

(137) In some implementations, a brake can be provided to stop the motion of the display unit **2202** about the axis **2206**, e.g., during deactivation of the display system. For example, a brake can be used which applies braking force about axis **2206** when power is removed so as to prevent rotation of display unit **2202**, e.g., during non-operation. In various implementations, the brake can include a rotary brake that is coupled to the shaft of motor **2218** or coupled to capstan pulley **2222** to reduce or prevent the motor shaft from rotating, and/or the brake can include a linear brake that is positioned between tilt member **2204** and display unit **2202**, and/or between tilt member **2204** and capstan drum **2216**/curved track **2210**, to cause friction in the relative movement of these components. For example, the brake can include a spring-loaded disc brake.

(138) One or more sensors can be coupled to one or more of the components of the display unit mechanism **2200** which are operative to detect the rotation of the display unit **2202** about axis **2206**. For example, in some implementations, a sensor **2230** (e.g., rotary encoder) is coupled to the shaft of motor **2218** and determines a rotational orientation of the motor shaft in order to detect the rotary orientation of the display unit **2202** about axis **2206**. In some implementations, a sensor can be coupled to other components of the display unit mechanism **2200** to sense the rotary orientation of the display unit **2202**. The sensor(s) can send signals describing sensed positions, orientations, or motion to one or more control circuits, e.g., a control system of the teleoperated system **100**. In some modes or implementations, the control circuit can provide control signals indicating the

sensed positions, orientations, or motion to a manipulator system. The sensors can be any of a variety of types of sensors, e.g., a rotary encoder, a magnetic sensor (e.g., magnetic incremental linear position sensor, Hall Effect sensor, etc.), optical sensor, encoder, resistance sensor, etc.

(139) In other example implementations, curved track **2210** and capstan drum **2216** can be coupled to tilt member **2204**, and groove member **2212** and motor **2218** can be coupled to display unit **2202**.

(140) FIG. **23** is a perspective view, FIG. **24** is a front view, and FIG. **25** is a side view of an example of another implementation of a display system **2300**, according to some implementations. In some examples, display system **2300** can be used in a user control system **102** of a teleoperated system as described for FIG. **1** and elsewhere herein, or can be used in other systems or as a standalone system as described above. Features of display system **2300** can be similar to those of display system **300** of FIG. **3** as described above, unless otherwise indicated.

(141) Display system **2300** includes a base support **2302**, an arm support **2304**, and a display unit **2306**. As described in greater detail below, display unit **2306** is provided with multiple degrees of freedom of movement by a support linkage including the base support **2302**, arm support **2304** coupled to the base support **2302**, and a track mechanism including a track member and a sliding member (as described below) coupled to the arm support **2304**. Display unit is coupled to the sliding member.

(142) In some implementations, base support **2302** and arm support **2304** can be similarly implemented as described above with reference to FIGS. **3-7**. In some examples, base support **2302** is a vertical member that is mechanically grounded, e.g., coupled to ground. Base support **2302** can be mechanically coupled to a support structure **2310**. Base support **2302** includes a first base portion **2312** and a second base portion **2314**. The first base portion **2312** is a proximal portion of the base support **2302** that can be mechanically grounded, and the second base portion **2314** is a distal portion of the base support **2302** that is linearly coupled to the first base portion **2312** such that the second base portion **2314** is translatable with respect to the first base portion **2312** in a linear degree of freedom. In some examples, first base portion **2312** and second base portion **2314** are telescopically coupled, e.g., first base portion **2312** is a first telescoping base portion and second base portion **2314** is a second telescoping base portion, such that one of the portions **2312** or **2314** is configured as a tube or sleeve with a hollow interior through which the other of the portions **2314** or **2312** extends. In the example of FIGS. **23-25**, second base portion **2314** is linearly translatable through an interior of first base portion **2312** in a linear degree of freedom **2316**. The linear translation of second base portion **2314** with respect to first base portion **2312** can be driven by one or more actuators, e.g., motors, e.g., similarly as described above with reference to FIGS. **15** and **16**. Other implementations can use different configurations. For example, first base portion **2312** can extend through the interior of second base portion **2314** such that second base portion **2314** can be linearly translated with respect to first base portion **2312**. In other examples, the base portions **2312** and **2314** can be positioned adjacent to each other along their vertical lengths to allow the linear translation.

(143) Arm support **2304** is a horizontal member that is mechanically coupled to the base support **2302**. Arm support **2304** includes a first arm portion **2318** and a second arm portion **2320**. The first arm portion **2318** is a proximal portion of the arm support **2304** that is rigidly coupled to the second base portion **2314** of base support **2302**, and the second arm portion **2320** is a distal portion of the arm support **2304** that linearly coupled to the first arm portion **2318** such that the second arm portion **2320** is linearly translatable with respect to the first arm portion **2318** in a linear degree of freedom. In some examples, first arm portion **2318** and second arm portion **2320** are telescopically coupled, e.g., first arm portion **2318** is a first telescoping arm portion and second arm portion **2320** is a second telescoping arm portion, such that one of the portions **2318** or **2320** is configured as a tube or sleeve with a hollow interior through which the other of the portions **2320** or **2318** extends. In the example of FIGS. **23-25**, second arm portion **2320** is linearly translatable through an interior

of first arm portion **2318** in a linear degree of freedom **2322**. The linear translation of the second arm portion **2320** with respect to the first arm portion **2318** can be driven by one or more actuators, e.g., motors, as described in greater detail with respect to FIGS. **17** and **18**. Other implementations can use different configurations, e.g., first arm portion **2318** can extend through the interior of second arm portion **2320** such that the second arm portion **2320** can be linearly translated with respect to the first arm portion **2318**. In other examples, the arm portions **2318** and **2320** can be positioned adjacent to each other along their vertical lengths to allow the linear translation.
(144) In some implementations, first arm portion **2318** and second base portion **2314** can be considered to be a single piece, e.g., a middle support or middle portion that is coupled between first base portion **2312** and second arm portion **2320**. The middle support includes the horizontal first arm portion **2318** coupled rigidly to the vertical second base portion **2314** that are oriented orthogonally to each other. Second arm portion **2320** is horizontally translatable in the degree of freedom **2322** with respect to the middle support, and the middle support and second arm portion **2320** are vertically translatable in the degree of freedom **2316** with respect to first base portion **2312**.
(145) In some examples as shown, arm support **2304** extends along a horizontal axis that is orthogonal to a vertical axis along which base support **2302** extends. In some examples, base support **2302** and arm support **2304** are fixed in orientation with respect to each other, e.g., they translate but do not change orientation with respect to each other. In some examples, the arm support **2304** extends along an axis above a user operating the display unit, and a vertical axis extending through the base support **2302** extends through the first arm portion **2318** of the arm support **2302**. In other implementations, arm support **2304** can extend at other heights and/or configurations, e.g., below a user's head or body, at the height of the user's head, in back of user and yoking around a user, etc. Some implementations can provide components in display system **2300** to reduce vibration in the supports and members of the display system **2300** to provide a smoother experience for a user operating the display unit **2306**, similarly as described for FIGS. **3-5**.
(146) Display unit **2306** is mechanically coupled to arm support **2304**. Display unit **2306** is moveable in two linear degrees of freedom provided by the linear translation of the second base portion **2314** and second arm portion **2320**. In some implementations, these linear degrees of freedom can be provided within a vertical plane. In some examples, as shown, the vertical plane can be defined by the base support **2302** and arm support **2304**.
(147) Display unit **2306** includes a display device that can display images, e.g., one or more display screens, projectors, or other devices. In some implementations, as in FIGS. **23-25**, the display unit **2306** includes two viewports **2323**, and the display device is provided behind or included in the viewports. In some implementations, one or more display screens or other display devices can be positioned on the display unit **2306** in place of viewports **2323**.
(148) Display unit **2306** is coupled to the arm support **2304** via a track member **2324** and a sliding member **2330**. In the example of FIGS. **23-25**, a first end of track member **2324** is rigidly coupled to a distal end of the second arm portion **2320**. The track member **2324** extends in a curved or bent configuration from its first end to a second end. For example, in the orientation of FIGS. **23-25**, the second end of the track member **2324** is lower than the first end. In this example, the track member **2324** extends approximately down (toward support structure **2310**) and toward the base support **2302** from the first end of the track member. In some implementations track member **2324** is curved about a central axis (e.g., axis **2326**). In some implementations, track member **2324** includes no linear segments or portions.
(149) Display unit **2306** is coupled to sliding member **2330**, and sliding member **2330** is slidably coupled to track member **2324**. In some implementations, as shown in FIGS. **23-25**, sliding member **2330** includes an aperture **2329** (or slot or groove) that extends through the sliding member, and track member **2324** extends through aperture **2329** to allow sliding member **2330** and

display unit **2306** to slide along the length of the track member. Sliding member **2330** and display unit **2306** can be positioned at any of multiple positions along track member **2324**. Track member **2324** thus provides a set of positions for the display unit **2306** that follow the curved configuration of track member **2324**.

(150) In the example of FIGS. **23-25**, track member **2324** extends along a curved path about an axis **2326**, which is a tilt axis for the display unit **2306**. In some implementations, a radius of the curved track member **2324** determines the position of the tilt axis **2326**. For example, the tilt axis **2326** can extend through a point that is a radius distance from the track member **2324**, the point being at the center of a circle (or other shape) at least partially traced by the track member **2324**. For example, as sliding member **2330** and display unit **2306** are moved along the track member **2324**, these elements move rotationally about tilt axis **2326** with respect to the second arm portion **2320** in a rotary (tilt) degree of freedom **2327**. In some implementations, tilt axis **2326** is oriented orthogonally to the linear degrees of freedom provided to the display unit **2306** by base support **2302** and arm support **2304**. For example, sliding member **2330** can provide a rotary degree of freedom to display unit **2306** that is in a vertical plane that is the same as, or is parallel to, the vertical plane in which the degrees of freedom **2316** and **2322** are provided by base support **2302** and arm support **2304**. In some implementations, tilt axis **2326** is orthogonal to the plane defined by the degrees of freedom **2316** and **2322**. In some implementations, the base support **2302**, arm support **2304**, and guiding member **2330** can be considered to be a support linkage having display unit **2306** coupled at the distal end of the support linkage.

(151) Sliding element **2330** and display unit **2306** can be rotated about a defined pivot axis based on movement allowed by first arm portion **2314**, second arm portion **2320**, and track member **2324**. Some examples of such rotation about defined pivot axes is described below with respect to FIGS. **26-31**.

(152) In some other example implementations, a different mechanism can be used to slidably or moveably couple display unit **2306** to track member **2324**. For example, a cam roller can be coupled to display unit **2306**, and track member **2324** can be a cam follower. The cam roller engages the cam follower. For example, the cam roller can be rotatably coupled to display unit **2306** and, in some implementations, have an axis of rotation that is orthogonal to yaw axis **2332**. The cam roller can be cylindrical and can roll along the curved surface of the cam follower to provide rotational motion about tilt axis **2326**. For example, the cam roller can be held against the cam follower by walls or ridges of the cam follower. In some implementations, the cam follower (track member **2324**) can be rigidly coupled to display unit **2306**, and the cam roller can be rotatably coupled to tilt member **2324**, such that the cam follower moves about tilt axis **2326** with display unit **2306**.

(153) The rotational motion of the display unit **2306** about tilt axis **2326** can be driven by one or more actuators, e.g., motors. In some implementations, a rotary motor **2328** can be rigidly coupled to the sliding member **2330**, and a rotating shaft of the motor **2328** can be coupled to a capstan pulley (e.g., similar to capstan pulley **2222** shown in FIG. **22**). A cable can be coupled between the first end and the second end of track member **2324** and run along the side of the track member **2324** facing motor **2328** and base support **2302** such that the track member **2324** functions as a capstan drum. The cable is wrapped around the capstan pulley that is connected to the motor shaft (e.g., similar to capstan drive mechanism described with respect to FIG. **22**). In some implementations, two cables can be used instead of the cable, e.g., a first cable having a first end attached to a first end of track member **2324** first end and a second end attached to the capstan pulley, and a second cable having a first end attached to a second end of track member **2324** and a second end attached to the capstan pulley, where the two cables are routed along the side surface of the track member **2324**.

(154) With this capstan drive mechanism, the motor **2328** can be controlled to rotate the capstan pulley in either direction and move the cable(s), which pulls display unit **2306** along track member

2324 in a corresponding direction. The motor **2328** can be controlled by control signals from a control circuit (e.g., control system) to move display unit **2306** about tilt axis **2326** to a particular orientation in the tilt degree of freedom **2327**. Other implementations can use different drive mechanisms to move display unit **2306** along track member **2324**.

(155) In some implementations, display unit **2306** is rotationally coupled to the sliding member **2330** and can be rotated with respect to the sliding member **2330** (and with respect to the track member **2324**, arm support **2304**, and base support **2302**) about a yaw axis **2332**. For example, this can be lateral or left-right rotation from the point of view of a user viewing images of display unit **2306** via viewports **2323**. In the example of FIGS. **23-25**, display unit **2306** is coupled to the sliding member **2330** by a rotary mechanism which can be a track mechanism. For example, in some implementations, the track mechanism includes a curved track bearing that includes a curved track **2334**, and curved track **2334** is coupled to display unit **2306** and the curved track **2334** slidably engages a groove member that is rigidly coupled to the sliding member **2330**, e.g., similarly operating as described above for FIGS. **3** and **22**. This allows the display unit **2306** to rotate about yaw axis **2332** in a rotary (yaw) degree of freedom **2333** by moving the curved track **2334** through a groove of the groove member. In some implementations, a curved track is coupled to sliding member **2330** and a groove member is coupled to display unit **2306**, and the groove member engages and slides along the length of the curved track to allow the rotary movement of the display unit **2306** about yaw axis **2332**. In some implementations, the groove member can be about as long as the width of sliding member **2330** and/or include at least a portion of a loop through which the curved track **2334** slides.

(156) In some implementations, curved track **2334** is a curved rail that slidably engages a groove member as described. In some implementations, a different mechanism can be used. In some examples, curved track **2334** can be a curved cam follower that engages a cam roller. For example, the cam roller can be rotatably coupled to display unit **2306** and, in various implementations, have an axis of rotation that is orthogonal to the yaw axis **2332**, or parallel to the yaw axis **2332**. For example, the cam roller can be cylindrical and can roll along the curved surface of the cam follower that is rigidly coupled to sliding member **2330**. For example, the cam roller can be held against the cam follower by walls or ridges of the cam follower. In some implementations, the cam follower can be rigidly coupled to the display unit **2306**, and the cam roller can be rotatably coupled to sliding member **2330**.

(157) The curvature (e.g., radius) of the curved track **2334** and/or groove member is selected to provide the yaw axis **2332** at a particular distance from a user-facing side of the display unit **2306** and/or from tilt axis **2326**. For example, the yaw axis **2332** can be provided at a horizontal distance (parallel to horizontal degree of freedom **2322**) from display unit **306** such that it approximately intersects a defined (virtual or software-defined) neck pivot axis corresponding to a pivot axis in a user's neck, as described below. The defined neck pivot axis can be used as a reference for motion of the display unit **2306** in some implementations. In the described implementation, the angle between yaw axis **2332** and a vertical axis (e.g., parallel to degree of freedom **2316**) varies based on the orientation of the display unit **2306** about tilt axis **2326**.

(158) The yaw motion of the display unit **2306** about yaw axis **2332** can be driven by one or more actuators, e.g., motors. For example, motor **2340** can be a rotary motor rigidly coupled to the sliding member **2330** and having a rotatable shaft that outputs force on the display unit about yaw axis **2332** using a drive transmission. In some examples, the drive transmission can include a capstan drive mechanism, e.g., similar to the capstan drive mechanism described above with reference to FIG. **22**. For example, the driven shaft of the motor **2340** can be coupled to a capstan pulley **2342**. A cable **2344** can be attached to both ends of a capstan drum **2336** that is rigidly coupled to curved track **2334**, where the cable is wrapped around the capstan pulley **2342**, or two cables can be attached between the capstan pulley **2342** and the respective ends of capstan drum **2336**. The motor **2340** rotates the capstan pulley **2342** to move the cable(s) (e.g., wind and unwind

the cable(s) on the pulley), thus rotating capstan drum **2336**, curved track **2334**, and display unit **2306** about yaw axis **2332**. In some implementations, other transmissions and/or couplings can be used to provide rotational motion of display unit **2306** about yaw axis **2332** with respect to the track member **2324** and arm support **2304**, e.g., using a rotary joint similarly as described above for display system **300**.

(159) Display system **2300** thus provides display unit **2306** with vertical linear degree of freedom **2316**, horizontal linear degree of freedom **2322**, rotational tilt degree of freedom **2327**, and rotational yaw degree of freedom **2333**. For example, the vertical and horizontal degrees of freedom allow the display unit **2306** to be moved to any position within a range of motion or allowed workspace (e.g., within a vertical plane), and the tilt degree of freedom allows the display unit to be moved to a particular orientation within its range of motion (e.g., within the vertical plane or a parallel vertical plane).

(160) A combination of coordinated movement of components of display system **2300** in at least two of these degrees of freedom allow the display unit **2306** to be positioned at various positions and orientations in its workspace, e.g., translated or rotated around a user, to facilitate a custom viewing experience for the user using the display unit. The motion of display unit **2306** in the tilt, horizontal, and/or vertical degrees of freedom allows display unit **2306** to stay close to the user's head and eyes during user head motion, and/or maintain a physical connection between the user's forehead and display unit **2306**.

(161) For example, display unit **2306** is positionable (e.g., translatable and/or rotatable) in its workspace such that eyes of the user align with the viewports of the display unit. In addition, display unit **2306** can be rotated in physical space about a defined eye pivot axis corresponding to (e.g., coincident with) an eye axis through both of a user's eyes to allow a desired vertical (e.g., up-down) eye viewing angle for the user. The display unit can be rotated about yaw axis **2332** to allow a desired yaw (e.g., left-right) viewing angle for the user. These rotations allow the display unit **2306** to be oriented comfortably for the user to view images through the viewports.

(162) The degrees of freedom also or alternatively allow the display system **2300** to provide motion of display unit **2306** in physical space about a different defined pivot axis that can be positioned in any of various locations in the workspace of the display unit **2306**. For example, the system **2300** can provide motion of display unit **2306** in physical space that corresponds to motion of a user's head when operating the display system **2300**. This motion can include rotation about a defined neck pivot axis that approximately corresponds to a neck axis of a user's head at the user's neck. This rotation allows the display unit **2306** to be moved in accordance with the user's head that is directing movement of the display unit **2306**, e.g., using a head input device similar to head input device **342** shown in FIGS. **3-5**. Some examples of such motion of a display unit about a neck pivot axis (FIGS. **26-29**) and an eye pivot axis (FIGS. **30-31**) are described below.

(163) In another example, the motion of display unit **2306** can include rotation about a defined forehead pivot axis that approximately corresponds to a forehead axis extending through the user's head at the user's forehead when the display unit **2306** is oriented, as shown, in a centered yaw rotary orientation about yaw axis **2332**. In some implementations, the forehead pivot axis corresponds to a forehead axis extending through a portion of an input device of the display unit **2306** (e.g., a head input device similar to head input device **342** of FIGS. **3-5**), where the portion is at or near a point of contact between the user's forehead and the input device. In some implementations, the defined forehead pivot axis can be oriented parallel to tilt axis **2326**, e.g., orthogonal to the linear degrees of freedom **2316** and **2322**. The defined forehead pivot axis can alternately be positioned to correspond to a different location or portion of the user's forehead or display unit.

(164) In another example, the motion of display unit **2306** can include rotation about a defined hand input device pivot axis that approximately corresponds to an axis that extends through one or more hand input controls of display unit **2306**. For example, this axis can extend through portions

(e.g., the centers of grips) of both hand input controls **2340***a* and **2340***b* that are positioned on opposing (left and right) sides of display unit **2306**. In some implementations, the defined hand input device pivot axis can be oriented parallel to tilt axis **2326**, e.g., orthogonal to the linear degrees of freedom **2316** and **2322**, similarly to the neck, eye, and forehead pivot axes described above. The hand input device pivot axis can alternately be positioned to correspond to a different location or portion of the display unit, e.g., a location of a different hand input device. For example, the hand input device pivot axis provides an axis of rotation about which the display unit **2306** can be commanded to be rotated by manipulation of the hand input devices **2340***a* and **2340***b* by the user, and/or by user manipulation of other user input devices such as head input device **342**, control input devices **210** and **212**, etc.

(165) In another example, the motion of display unit **2306** in its workspace can include linear motion, e.g., based on linear translation in the vertical linear degree of freedom **2316** and horizontal linear degree of freedom **2322** and without rotational motion in the tilt and/or yaw degrees of freedom **2327** and/or **2333**. In another example, the motion can include both linear motion and rotational motion. For example, display unit **2306** can be moved linearly and then rotationally in its workspace, and/or vice-versa.

(166) During rotation of the display unit **2306** about a defined pivot axis, the movement of base support **2302** in vertical degree of freedom **2316** and/or the movement of arm support **2304** in horizontal degree of freedom **2322** can have a change in direction without a change in direction in rotation of the display unit about the defined pivot axis. For example, arm support **2304** can reverse direction as the display unit is rotated from fully up to fully down orientations. Thus, it is possible in some implementations for the base support or arm support to end up in the same position while the display unit has rotated from a first orientation to a second orientation about the defined pivot axis, where the support moves back and forth while the display unit transitions between those two orientations.

(167) Display unit **2306** can include input devices to allow a user to provide input to manipulate the orientation and/or position of the display unit **2306** in space, and/or to manipulate other functions or components of the display system **2300** and/or larger system (e.g., teleoperated system). Some examples of input devices are described with respect to FIGS. **3-5** (e.g., hand input devices **340** and head input device **342**), and such input devices can be used in display system **2300**, e.g., provided on the sides and/or front facing surface of display unit **2306** in a similar manner. For example, hand input devices **2340***a* and **2340***b* can be located on display unit **2306** similarly to hand input devices **340***a* and **340***b*. User input provided from such input devices can be used to control motion of the display unit **2306** and/or displayed images and other components similarly as described above with reference to FIGS. **3-14**.

(168) In some implementations of a display system, display unit **2306** is rotatable about yaw axis **2332** in degree of freedom **2333** and one or more of the other degrees of freedom **2316**, **2322**, and/or **2327** are omitted from the display system **2300**. For example, display unit **2306** can be rotated about yaw axis **2332** (e.g., by actuator(s) and/or manually by a user) and the display unit **2306** can be manually positioned higher and/or lower (e.g., by actuator(s) and/or manually by a user), e.g., using base support **2302** or other mechanism, and horizontal degree of freedom **2322** and/or tilt degree of freedom **2327** are omitted.

(169) FIGS. **26-29** are side views of a portion of a display system **2600** showing rotation of a display unit about an example defined neck pivot axis, according to some implementations. Display system **2600** includes similar components to the display system **2300** of FIGS. **23-25** described above.

(170) In FIG. **26**, display system **2600** is shown with a display unit in a first pivot orientation. Display system **2600** includes a base support **2602**, an arm support **2604**, and a display unit **2606**. In this example, similarly to examples described above, base support **2602** includes a first base portion **2612** that is grounded and a second base portion **2614**. Second base portion **2614** is linearly

translatable through an interior of first base portion **2612** in a linear degree of freedom **2616**. Arm support **2604** includes a first arm portion **2618** rigidly coupled to second base portion **2614**, and a second arm portion **2620**. Second arm portion **2620** is linearly translatable through an interior of first arm portion **2618** in a linear degree of freedom **2622**.

(171) Display unit **2606** is coupled to the second arm portion **2620** via a track member **2624** and a sliding member **2630** similarly as described in examples above with respect to FIGS. **23-25**. Track member **2624** is rigidly coupled to the second arm portion **2620**. Display unit **2606** is coupled to sliding member **2630**, and sliding member **2630** is slidably coupled to track member **2624**. Display unit **2606** rotates about tilt axis **2626** in accordance with the movement of sliding member **2630** along track member **2624**.

(172) In FIG. **26**, display unit **2606** is oriented at the first pivot orientation about a defined neck pivot axis **2640** that can be approximately horizontal and can extend approximately parallel to tilt axis **2626**, e.g., orthogonal to a plane defined by the degrees of freedom **2616** and **2622**. In the example implementation, neck pivot axis **2640** can be positioned to intersect a head or neck of a user **2650** of display unit **2606**. For example, the neck pivot axis **2640** is positioned at a location that is aligned (e.g., approximately aligned) with a pivot axis of a neck of a typical user such as user **2650** when the display unit **2606** is oriented, as shown, in a centered yaw rotary orientation about yaw axis **2632**. For example, the pivot axis of the neck is a location at which a neck of a typical user operating the display unit approximately pivots. This location may be different in different users (e.g., users of different heights, different sizes of necks, etc.), and/or users may have different preferences as to where the defined neck pivot axis is located, such that in some implementations, a location determined (e.g., averaged) from preferred pivot locations of multiple users can be used for the defined neck pivot axis. In some example implementations, the defined neck pivot axis can be located at a location that is approximately aligned with a particular bone or any particular cervical vertebra of a neck of a user operating the display unit, e.g., an atlas bone or axis bone of a neck. In further examples, the defined neck pivot axis can be defined to correspond to other locations of a user's neck.

(173) The neck pivot axis **2640** is a virtual axis that is defined based on system parameters. The movement of the base support **2602**, arm support **2604**, and sliding member **2630** can provide the rotation about neck pivot axis **2640**. Neck pivot axis **2640** can be at different positions than shown in FIG. **26**, as defined by the movement of these components in creating rotation about the desired neck pivot axis. Some implementations may allow the location of the defined neck pivot axis to be changed, e.g., by user input, to accommodate the specific preferences or physiology of particular users. In some examples, the defined neck pivot axis can be changed to a different location at which the defined neck pivot axis is parallel to tilt axis **2626** (such as a different location in which the axis is orthogonal to a vertical plane defined by the degrees of freedom **2616** and **2622**), e.g., a new location that is up, down, forward, and/or back from a previous location in the point of view of a user. In some implementations, a stored profile or settings can be stored in association with a particular user and applied for use of the display system by that user, where the profile includes preferred locations for the defined neck pivot axis and/or defined eye pivot axis (described below), any of which can be loaded for system operation. These and other features (e.g., guiding the user to determine a defined neck pivot axis) can be similar to those described with respect to FIG. **8**.

(174) In the first pivot orientation shown in FIG. **26**, for example, display unit **2606** can be in a horizontal view orientation about neck pivot axis **2640**. For example, a view orientation of display unit **2606** with respect to neck pivot axis **2640** is indicated by a line **2642** that, in FIG. **26**, extends horizontally through the neck pivot axis **2640**. Rotation of display unit **2606** about the neck pivot axis **2640** is achieved by moving portion **2614** of the base support **2602**, portion **2620** of the arm support **2604**, sliding member **2630** to appropriate positions and/or orientations.

(175) FIG. **26** also shows a defined eye pivot axis **2660**. The defined eye pivot axis **2660** is positioned at a location that corresponds to (e.g., is coincident with) an eye axis that intersects the

eyes of a typical user such as user **2650** that is looking through the viewports of the display unit **2606** when the display unit **2606** is oriented, as shown, in a centered yaw rotary orientation about yaw axis **2632**. In some implementations, the defined eye pivot axis extends approximately parallel to tilt axis **2626**, e.g., orthogonal to a plane defined by the degrees of freedom **2616** and **2622**. This defined eye pivot axis can be orthogonal to the view orientation of the display unit **2606** when the display unit **2606** is oriented, as shown, in a centered yaw rotary orientation about axis **2632**. A view orientation of display unit **2606** with respect to eye pivot axis **2660** is indicated by a sight line **2662** that, in FIG. **26**, extends horizontally through eye pivot axis **2660**. In some implementations, as shown, the tilt axis **2626** is located below the sight line **2662**. Movement of the display system **2600** based on the eye pivot axis **2660** is described in greater detail below with respect to FIGS. **31** and **32**. Movement of the display system **2600** about other defined pivot axes of the display unit **2606**, e.g., forehead pivot axis or hand input device axis, can be similarly implemented to the movement about the neck and/or eye pivot axes.
(176) In FIG. **27**, display unit **2606** is oriented at a second pivot orientation about the defined neck pivot axis **2640** to allow an "upward" view angle of the user **2650**. An angle A**1** is an example angle of rotation of the display unit **2606** about the neck pivot axis **2640**, where angle A**1** is the angle between the horizontal view orientation **2642** of the display unit **2606** and a neck-axis view orientation **2742** of display unit **2606**. Display unit **2606** has been rotated about the neck pivot axis **2640** relative to the first pivot orientation shown in FIG. **26** by moving the base support **2602**, arm support **2604**, and sliding member **2630** to positions and/or orientations as shown. For example, relative to the first pivot orientation shown in FIG. **26**, the display unit **2606** is moved to the second pivot orientation by moving the sliding tilt member **2630** upward along track member **2624** to the orientation shown (counterclockwise about axis **2626** in the viewpoint of FIG. **27**), linearly translating second arm portion **2620** away from base support **2602** and toward the user **2650** to the position shown, and linearly translating second base portion **2614** upward (e.g., away from ground) to the position shown. The tilt axis **2626** is independent of the neck pivot axis **2640**, since rotation of the display unit **2606** about the neck pivot axis **2640** uses rotation about the tilt axis **2626** as well as linear motion (translation) in the degrees of freedom **2622** and **2616**.
(177) In some implementations, the view angle shown in FIG. **27** is the upward (e.g., counterclockwise as shown in FIG. **27**) rotational limit of the display unit **2606**. In other implementations, display unit **2606** can be rotated upward (e.g., counterclockwise) by greater amounts than shown in FIG. **27** before reaching a limit to rotation in this direction.
(178) In FIG. **28**, display unit **2606** is oriented at a third pivot orientation about the defined neck pivot axis **2640** to allow a "downward" view angle of the user **2650**. An angle A**2** is an example angle of rotation of the display unit **2606** about the neck pivot axis **2640**, where angle A**2** is the angle between the horizontal view orientation **2642** of the display unit **2606** and a neck-axis view orientation **2842** of the display unit. Display unit **2606** has been rotated about the neck pivot axis **2640** by moving the base support **2602**, arm support **2604**, and sliding member **2630** to positions and/or orientations as shown. For example, relative to the first pivot orientation shown in FIG. **26**, the display unit **2606** is moved to the third pivot orientation by moving the sliding member **2630** downward to the orientation shown (clockwise about axis **2626** in the viewpoint of FIG. **28**), linearly translating second arm portion **2620** in degree of freedom **2622** toward the user **2650** and away from base support **2602** to the position shown, and linearly translating second base portion **2614** downward (e.g., toward ground) to the position shown.
(179) In FIG. **29**, display unit **2606** is oriented at a fourth pivot orientation about the defined neck pivot axis **2640** to allow a further downward view angle of the user **2650**. An angle A**3** is an example angle of rotation of the display unit **2606** about the neck pivot axis **2640**, where angle A**3** is the angle between the horizontal view orientation **2642** of the display unit **2606** and a neck-axis view orientation **2942** of the display unit. Display unit **2606** has been rotated about the neck pivot axis **2640** by moving the base support **2602**, arm support **2604**, and sliding member **2630** to

positions and/or orientations as shown. For example, relative to the third pivot orientation shown in FIG. **28**, the display unit **2606** is moved to the fourth pivot orientation by moving the sliding member **2630** further downward to the orientation shown (clockwise about axis **2626** in the viewpoint of FIG. **29**), linearly translating second arm portion **2620** in degree of freedom **2622** toward the user **2650** and away from base support **2602** to the position shown, and linearly translating second base portion **2614** downward (e.g., toward ground) to the position shown.

(180) In the example implementations shown in FIGS. **26-29**, the distance between defined neck pivot axis **2640** and tilt axis **2626** is fixed during rotation of the display unit about the defined neck pivot axis **2640**.

(181) FIGS. **30** and **31** are side views of a portion of display system **2600** of FIG. **26** showing rotation of a display unit about a defined eye pivot axis, according to some implementations. Display system **2600** includes similar components to the display system **2300** described above.

(182) In FIG. **30**, display unit **2606** is oriented at a fifth pivot orientation about the defined eye pivot axis **2660** to allow an upward view angle of the user **2650**. An angle A**4** is an example angle of rotation of the display unit **2606** about the eye pivot axis **2660**, where angle A**4** is the angle between the horizontal eye-axis view orientation **2662** of the display unit **2606** (as shown in FIG. **26**) and an eye-axis view orientation **3062** of the display unit. Display unit **2606** has been rotated about the eye pivot axis **2660** relative to the first pivot orientation shown in FIG. **26** by moving the base support **2602**, arm support **2604**, and sliding member **2630** to positions and/or orientations as shown. For example, relative to the first pivot orientation shown in FIG. **26**, the display unit **2606** is moved to the fifth pivot orientation by moving sliding member **2630** upward to the angle shown (counterclockwise about axis **2626** in the viewpoint of FIG. **30**), linearly translating second arm portion **2620** toward base support **2602** and away from the user **2650** to the position shown, and linearly translating second base portion **2614** downward (e.g., toward ground) to the position shown (this movement is very small in this example). The tilt axis **2626** is independent of the eye pivot axis **2660**, since rotation of the display unit **2606** about the eye pivot axis **2660** may use rotation about the tilt axis **2626** as well as linear motion (translation) in the degrees of freedom **2622** and **2616**.

(183) In FIG. **31**, display unit **2606** is oriented at a sixth pivot orientation about the defined eye pivot axis **2660** to allow a downward view angle of the user **2650**. An angle A**5** is an example angle of rotation of the display unit **2606** about the eye pivot axis **2660**, where angle A**5** is the angle between the horizontal eye-axis view orientation **2662** of the display unit **2606** and an eye-axis view orientation **3162** of the display unit. Display unit **2606** has been rotated about the eye pivot axis **2660** by moving the base support **2602**, arm support **2604**, and member **2630** to positions and/or orientations as shown. For example, relative to the first pivot orientation shown in FIG. **26**, the display unit **2606** is moved to the sixth pivot orientation by moving sliding member **2630** downward to the orientation shown (clockwise about axis **2626** in the viewpoint of FIG. **31**), linearly translating second arm portion **2620** in degree of freedom **2622** away from base support **2602** and toward the user **2650** to the position shown, and linearly translating second base portion **2614** upward (e.g., away from ground) to the position shown.

(184) In this example implementation of FIGS. **30** and **31**, the eye pivot axis **2660** is close in position to the tilt axis **2626**. Thus, rotation about eye pivot axis **2660** is mostly provided by rotation about tilt axis **2626** and does not require substantial movement of second arm portion **2620** and second base portion **2614**. Other implementations may have a larger distance between eye pivot axis **2660** and tilt axis **2626** and require larger motions of second arm portion **2604** and second base portion **2614**.

(185) Display system **2600** (and **2300**) can provide movement of a display unit about a neck pivot axis and/or eye pivot axis of a user similar to the movement shown above in FIGS. **8-14** for the display systems **300**, **600** and **700** described above. In some implementations for particular defined pivot axes, display system **2600** (and **2300**) can provide such movement with overall less motion

and/or movement range required of the moving portions of arm support **2604** and base support **2602** than required from display systems **300**, **600**, and/or **700**. In these examples, this is due to the tilt axis **2626** of the display unit **2606** being closer to the neck pivot axis **2640** and to the eye pivot axis **2660**, allowing rotation of the display unit **2606** about tilt axis **2626** that is closer to the defined neck pivot axis or defined eye pivot axis.

(186) FIG. **32** is a perspective view of another implementation **3200** of a display system, and FIG. **33** is a front view, FIG. **34** is a side view, FIG. **35** is a top view, and FIG. **36** is a bottom view of a portion **3201** of the display system **3200**, according to some implementations. In some examples, display system **3200** can be used in a user control system **102** of a teleoperated system as described for FIG. **1** and elsewhere herein, or can be used in other systems or as a standalone system as described above. Features of display system **3200** can be similar to those of display system **300** of FIG. **3** and/or display system **2300** as described above, unless otherwise indicated.

(187) Display system **3200** includes a base support **3202**, an arm support **3204**, and a display unit **3206**. Display unit **3206** is provided with multiple degrees of freedom of movement by a support linkage including the base support **3202** and arm support **3204** coupled to the base support **3202**, where the display unit **3206** is coupled to the arm support **3204**. In some implementations, base support **3202** and the proximal portion of arm support **3204** (coupled to base support **3202**) can be similarly implemented as described above with reference to FIGS. **3-7** and **23-25**.

(188) In some examples, base support **3202** is a vertical member that is mechanically grounded, e.g., similarly as in implementations of FIGS. **3-7** and **23-25**, and includes a first base portion **3212** and a second base portion **3214**. The first base portion **3212** and second base portion **3214** can be linearly coupled (e.g., telescopically coupled) and allow linear translation of second base portion **3214** with respect to first base portion **3212** in a linear degree of freedom **3216**, where such translation can be driven by one or more actuators, e.g., motors, e.g., similarly as described above with reference to FIGS. **15** and **16**. Other implementations can use different configurations similarly as described above for implementations of FIGS. **3-7** and **23-25**.

(189) Arm support **3204** is a horizontal member that is mechanically coupled to the base support **3202**. Arm support **3204** includes a first arm portion **3218** and a second arm portion **3220**. First arm portion **3218** is a proximal portion of the arm support **3204** that is rigidly coupled to the second base portion **3214** of base support **3202**, and the second arm portion **3220** is a distal portion of the arm support **3204** that linearly coupled to the first arm portion **3218** such that the second arm portion **3220** is linearly translatable with respect to the first arm portion **3218** in a linear degree of freedom **3222**. In some examples, a proximal portion **3224** of second arm portion **3220** is telescopically coupled to first arm portion **3218**, e.g., the portions **3218** or **3220** are telescoping portions similarly as described in implementations of FIGS. **2-7** and **23-25**. In the example of FIGS. **32-36**, proximal portion **3224** of second arm portion **3220** is linearly translatable through an interior of first arm portion **3218** in linear degree of freedom **3222**. The linear translation of the second arm portion **3220** with respect to the first arm portion **3218** can be driven by one or more actuators, e.g., motors, some examples of which are described with respect to FIGS. **17** and **18**. Other implementations can use different configurations similarly as described above for implementations of FIGS. **3-7** and **23-25**. In some implementations, first arm portion **3218** and second base portion **3214** can be considered to be a single piece, e.g., a middle support or middle portion that is coupled between first base portion **3212** and second arm portion **3220**, similarly as described above.

(190) In some examples as shown, arm support **3204** extends along a horizontal axis that is orthogonal to a vertical axis along which base support **3202** extends. In some examples, base support **3202** and arm support **3204** are fixed in orientation with respect to each other, e.g., they translate but do not change orientation with respect to each other. In some examples, a vertical axis extending through the base support **3202** extends through the first arm portion **3218** of the arm support **3204**. In various implementations, arm support **3204** can extend at various heights and/or

configurations, e.g., below a user's head or body, at the height of the user's head, above a user's head, etc. Some implementations can provide components in display system **3200** to reduce vibration in the supports and members of the display system **3200**, similarly as described for FIGS. **3-5**.

(191) Second arm portion **3220** includes proximal portion **3224** that is coupled to the first arm portion **3202** as described above, and a yoke portion **3226** that is rigidly coupled to the proximal portion **3224**. In the described implementations, yoke portion **3226** includes two yoke members **3228***a* and **3228***b* (collectively referred to as **3228**) which extend within a horizontal plane that is orthogonal to a plane defined by the proximal portion **3224** and the base member **3202**. Yoke members **3228** extend in approximately opposing directions from the proximal portion **3224** and then in parallel directions parallel to the proximal portion **3224**, forming an approximate U-shape as shown. The yoke members may extend in various directions in other implementations, e.g., in an approximate V-shape or other shapes.

(192) Display unit **3206** includes a display device that can be similar as described for the implementations of FIGS. **3-7** and/or FIGS. **23-25**. Display unit **3206** is rotationally coupled to arm support **3204**. In these implementations, display unit **3206** is positioned between and rotationally coupled to the distal ends of the yoke members **3228**. In this example, display unit **3206** is coupled to a crossbeam **3244** that is rotatably coupled to the yoke members **3228**. For example, a shaft **3229***a* of crossbeam **3244** is rotatably coupled to yoke member **3228***a*, and a shaft **3229***b* of crossbeam **3244** is rotatably coupled to yoke member **3228***b*. As crossbeam **3244** rotates, display unit **3206** rotates with the crossbeam. In some implementations, shafts **3229** can be rigidly coupled to display unit **3206** without use of crossbeam **3244**, e.g., if yaw motion is not being implemented for display unit **3206**.

(193) Display unit **3206** is rotatable in a rotary (tilt) degree of freedom **3232** about a tilt axis **3230** with respect to arm support **3204** and base support **3202**. Tilt axis **3230** extends through the distal ends of yoke members **3228** and, for example, can be aligned with or parallel to the lengths of shafts **3229***a* and **3229***b*. In some implementations, tilt axis **3230** is oriented orthogonally to the plane defined by linear degrees of freedom **3216** and **3222** provided to the display unit **3206** by base support **3202** and arm support **3204**. For example, tilt degree of freedom **3232** can be provided in a vertical plane that is the same as, or is parallel to, the vertical plane in which the degrees of freedom **3216** and **3222** are provided by base support **3202** and arm support **3204**. In some implementations, base support **3202** and arm support **3204** can be considered to be a support linkage having display unit **3206** coupled at the distal end of the support linkage.

(194) Display unit **3206** is moveable in two linear degrees of freedom provided by the linear translation of the second base portion **3214** and second arm portion **3220**. For example, these linear degrees of freedom can be provided within a vertical plane. In some examples, as shown, the vertical plane can be defined by the base support **3202** and proximal portion **3224** of arm support **3204**.

(195) Display unit **3206** can be rotated about a defined pivot axis based on translation provided by first arm portion **3214** and second arm portion **3220**, and by rotation about tilt axis **3230** allowed by the rotational coupling of the display unit **3206** to the yoke portion **3226**. Rotation about a defined pivot axis can be similar to such rotation described above for implementations of FIGS. **3-7** and **23-25** and is further described with respect to FIG. **39**.

(196) The rotational motion of the display unit **3206** about tilt axis **3230** can be driven by one or more actuators, e.g., motors. In some implementations, a rotary motor **3236** (shown in FIGS. **33**, **35**, and **36** in dotted lines) can be rigidly coupled to one of the yoke members **3228** (or a respective motor coupled to each yoke member **3228**), and a rotating shaft of the motor **3236** can be coupled to crossbeam **3244** (e.g., one of shafts **3229**) that is coupled to the display unit **3206**. In some implementations, a drive mechanism can be coupled to motor **3236**, e.g., a gear mechanism or a capstan mechanism similar to capstan mechanism shown in FIG. **22**. In an example capstan

mechanism, a capstan drum is coupled to the driven shaft **3229** (or elsewhere on crossbeam **3244**) and a capstan pulley is coupled to the shaft of motor **3236**, with one or more cables coupled between drum and pulley. The motor **3236** can be controlled by control signals from a control circuit (e.g., control system) to move crossbeam **3244** and display unit **3206** about tilt axis **3230** to a particular orientation in the tilt degree of freedom **3232**.

(197) In some implementations, display unit **3206** is additionally rotatably coupled to the yoke portion **3226** of arm support **3204** to allow rotation of the display unit **3606** about a yaw axis **3240** in a yaw degree of freedom **3242** with respect to arm support **3204** and base support **3202**. For example, this can be lateral or left-right rotation from the point of view of a user viewing images of display unit **3206**, e.g., via viewports. In the example of FIGS. **32-36**, display unit **3206** is coupled to the yoke portion **3226** by a rotary mechanism which can be a track mechanism. In some implementations, the track mechanism can be similar to the track mechanisms described above in implementations that provide yaw axis movement. For example, the track mechanism can include crossbeam **3244** and a curved track bearing that includes a curved track **3246**. Curved track **3246** slidably engages a groove member **3248**, e.g., operating similarly as described above for FIGS. **3** and **22**. Display unit **3206** moves along a curved path constrained by the curved track **3246**, causing display unit **3206** to rotate about yaw axis **3240** in rotary degree of freedom **3242**. Additional examples are described below with respect to FIGS. **37** and **38**. For example, curved track **3246** can be rigidly coupled to crossbeam **3244** (which is rigidly coupled to yoke portion **3226** with respect to the yaw degree of freedom), and groove member **3248** can be rigidly coupled to display unit **3206**. Alternatively, a groove member can be rigidly coupled to crossbeam **3244** and the curved track can be rigidly coupled to display unit **3206**. Additional groove members can be provided along the length of the curved track **3246** in some implementations.

(198) In some implementations, a different mechanism can be used to provide rotation of display unit **3206** about yaw axis **3240**. In some examples, curved track **3246** can be a curved cam follower that engages a cam roller, similarly as described above for yaw movement implementations in FIGS. **3-5** and/or **23-25**.

(199) The curvature (e.g., radius) of the curved track **3246** and/or groove member **3248** is selected to provide the yaw axis **3240** at a particular distance from a user-facing side of the display unit **3206** and/or from tilt axis **3230**. For example, the yaw axis **3240** can be provided at a horizontal distance (parallel to horizontal degree of freedom **3222**) from display unit **3206** such that it approximately intersects a defined (virtual or software-defined) neck pivot axis corresponding to a pivot axis in a user's neck, as described below. The defined neck pivot axis can be used as a reference for motion of the display unit **3206** in some implementations. In the described implementation, the angle between yaw axis **3240** and a vertical axis (e.g., parallel to degree of freedom **3216**) varies based on the orientation of the display unit **3206** about tilt axis **3230**.

(200) The yaw motion of the display unit **3206** about yaw axis **3240** can be driven by one or more actuators, e.g., motors. For example, a rotary motor (not shown) can be rigidly coupled to the display unit **3206** and can include a rotatable shaft that outputs force on display unit **3206** about yaw axis **3240** using a drive transmission, e.g., a capstan drive mechanism, gear mechanism, etc. In some examples, the drive transmission can include a capstan drive mechanism, e.g., similar to the capstan drive mechanism described above with reference to FIG. **22**. For example, the driven shaft of the motor can be coupled to a capstan pulley and a cable can be attached to both ends of a capstan drum that is rigidly coupled to curved track **3246**, where the cable is wrapped around the capstan pulley (or two cables can be attached between the capstan pulley and the respective ends of capstan drum). The motor rotates the capstan pulley to move the cable(s) (e.g., wind and unwind the cable(s) on the pulley), thus rotating the capstan drum, curved track **3246**, and display unit **3206** about yaw axis **3240**. In some implementations, the motor and/or drive transmission can be at least partially located within a housing of display unit **3206**. For example, the capstan drum can be positioned on a side of track member **3244** or crossbeam **3244** that is opposite to the user position,

and the capstan pulley can be placed adjacent to the capstan drum further from the user position. In some implementations, other transmissions and/or couplings can be used to provide rotational motion of display unit **3206** about yaw axis **3240** with respect to the track member **3244** and arm support **3204**.

(201) Display system **3200** thus provides display unit **3206** with vertical linear degree of freedom **3216**, horizontal linear degree of freedom **3222**, rotational tilt degree of freedom **3232**, and rotational yaw degree of freedom **3242**. For example, the vertical and horizontal degrees of freedom allow the display unit **3206** to be moved to any position within a range of motion or allowed workspace (e.g., within a vertical plane), and the tilt degree of freedom allows the display unit to be moved to a particular orientation within its range of motion (e.g., within the vertical plane or a parallel vertical plane).

(202) A combination of coordinated movement of components of display system **3200** in at least two of these degrees of freedom allow the display unit **3206** to be positioned at various positions and orientations in its workspace, e.g., translated or rotated around a user, to facilitate a custom viewing experience for the user, similarly as described above for the implementations of FIGS. **3-7** and/or **23-25**. Further, the degrees of freedom of the display unit **3206** also or alternatively allow the display system **3200** to provide motion of display unit **3206** in physical space about a defined pivot axis that can be positioned in any of various locations in the workspace of the display unit **2306**, similarly as described above, e.g., an eye pivot axis, a forehead pivot axis, a neck pivot axis, etc.).

(203) Display unit **3206** can include input devices to allow a user to provide input to manipulate the orientation and/or position of the display unit **3206** in space, and/or to manipulate other functions or components of the display system **3200** and/or larger system (e.g., teleoperated system). Some examples of input devices are described with respect to FIGS. **3-5** (e.g., hand input devices **340** and head input device **342**), and such input devices can be used in display system **3200**, e.g., provided on the sides and/or front facing surface of display unit **3206** in a similar manner. User input provided from such input devices can be used to control motion of the display unit **3206** and/or displayed images and other components similarly as described above with reference to FIGS. **3-14**.

(204) In some implementations of a display system, display unit **3206** is rotatable about yaw axis **3240** in degree of freedom **3242** and one or more of the other degrees of freedom **3216**, **3222**, and/or **3232** are omitted from the display system **3200**. For example, display unit **3206** can be rotated about yaw axis **3240** (e.g., by actuator(s) and/or manually by a user) and the display unit **3206** can be manually positioned higher and/or lower (e.g., by actuator(s) and/or manually by a user), e.g., using base support **3202** or other mechanism, where horizontal degree of freedom **3222** and/or tilt degree of freedom **3232** are omitted.

(205) FIGS. **37** and **38** are perspective views of portion **3201** of the display system **3200** shown in FIG. **32**. A perspective view of the bottom of the yoke portion **3226** and display unit **3206** is shown.

(206) In FIG. **37**, display unit **3206** is shown at a centered yaw orientation about the yaw axis **3240**. For example, groove member **3248** is rigidly coupled to display unit **3206** and is centered along the curved length of the track member **3246**.

(207) In FIG. **38**, display unit **3206** is shown at an orientation about the yaw axis **3240** that is changed with respect to FIG. **37**. For example, groove member **3248** has been moved along the track member **3246**, causing display member **3206** to be moved left, e.g., clockwise about the yaw axis **3240** if viewed from the bottom of display unit **306**. As shown, track member **3246** can have a curved length that provides the display member **3206** with a limited rotational range within the space between the yoke members **3228**, such that the display member **3206** can rotate about yaw axis **3240** without impacting or contacting the yoke members **3228**. For example, to enable this limited rotational yaw range, the curved track can extend across (or be provided as) a central portion of crossbeam **3244**. In this example, the curved track does not extend outside of the central portion, preventing groove member **3248** from moving outside that central portion. In some

implementations, stops can be positioned at ends of the track member **3246** to prevent movement of groove member **3248** beyond the length of the track member **3246**.

(208) FIG. **39** is a side view of portion **3201** of display system **3200** of FIG. **32**, showing rotation of a display unit used by a user, according to some implementations. In FIG. **39**, display unit **3206** is in a pivot orientation about a defined neck pivot axis **3902** that can be approximately horizontal and can extend approximately parallel to tilt axis **3230**, e.g., orthogonal to a plane defined by the degrees of freedom **3216** and **3222**. In the example implementation, neck pivot axis **3230** can be positioned to intersect a head or neck of a user **3950** of display unit **3206**. For example, the neck pivot axis **3902** can be similar to neck pivot axis **840** or **2640** described above. The movement of the base support **3202**, arm support **3204**, and display unit **3206** about tilt axis **3230** can provide the rotation about neck pivot axis **3902**. Neck pivot axis **3920** can be at different positions than shown in FIG. **39**, as defined by the movement of these components in creating rotation about the desired neck pivot axis. These and other features (e.g., guiding the user to determine a defined neck pivot axis) can be similar to those described with respect to FIG. **8**.

(209) A view orientation of display unit **3206** with respect to neck pivot axis **3902** is indicated by a line **3904** that, in FIG. **39**, extends through the neck pivot axis **3902** and is oriented to provide a downward angle with reference to a horizontal line **3905** in this example. Rotation of display unit **3206** about the neck pivot axis **3902** is achieved by moving portion **3214** of the base support **2602**, portion **3220** of the arm support **2604**, and display unit **3206** to appropriate positions and/or orientations. This movement can be similar in principle to the movement described with respect to FIGS. **6-11** and/or **26-29**, although one or more movement directions and magnitudes may be different due to the different position of the tilt axis **3230**.

(210) FIG. **39** also shows a defined eye pivot axis **3906**. The defined eye pivot axis **3906** is positioned at a location that corresponds to (e.g., is coincident with) an eye axis that intersects the eyes of a typical user such as user **3950** that is looking through viewports or at a display device of the display unit **3206** when the display unit **3206** is oriented, as shown, in a centered yaw rotary orientation about yaw axis **3240**. In some implementations, the defined eye pivot axis extends approximately parallel to tilt axis **3230**, e.g., orthogonal to a plane defined by the degrees of freedom **3216** and **3222**. This defined eye pivot axis can be orthogonal to the view orientation of the display unit **3206** when the display unit **3206** is oriented, as shown, in a centered yaw rotary orientation about yaw axis **3240**. A view orientation of display unit **3206** with respect to eye pivot axis **3906** is indicated by a sight line **3908** that, in FIG. **26**, extends through eye pivot axis **3906** and is oriented to provide a downward view angle with reference to a horizontal sight line **3907** in this example. Movement of display unit **3906** about defined eye pivot axis **3906** can be similar in principle to the movement described with respect to FIGS. **12-14** and/or **30-31**, although one or more movement directions and magnitudes may be different due to the different position of the tilt axis **3230**. Movement of the display system **3200** about on other defined pivot axes of the display unit **3206**, e.g., forehead pivot axis or hand input device axis, can be similarly implemented to the movement about the neck and/or eye pivot axes.

(211) Display system **3200** can position the tilt axis **3230** closer to some defined pivot axes than in other implementations, e.g., implementations such as display system **300**, **600**, and **700** of FIGS. **3-7**. In some implementations for particular defined pivot axes, display system **3200** can provide movement of display unit **3206** with overall less motion and/or movement range required of the moving portions of arm support **3204** and base support **3202** than required from display systems **300**, **600**, and/or **700**. In these examples, this is due to the tilt axis **3230** of the display unit **3206** being closer to the neck pivot axis **3240** and to the eye pivot axis **3260**, allowing rotation of the display unit **3206** about tilt axis **3226** that is closer to the defined neck pivot axis or defined eye pivot axis.

(212) FIG. **40** is a perspective view of an example device portion **4000** of a control input device which can be used in one or more implementations described herein. In some implementations,

device portion **4000** can be used as a portion of control input device **210** or **212** as described above with reference to FIGS. **1** and **2**, which can be used with any of the various display system implementations described herein. In some implementations, the device portion **4000** includes one or more gimbal mechanisms.
(213) In some implementations, device portion **4000** can be a mechanically grounded controller. For example, the device portion **4000** can be coupled to a mechanical linkage that is coupled to the ground or an object connected to ground, providing a stable platform for the use of the device portion **4000**.
(214) Device portion **4000** includes a handle **4002** which is contacted by a user to manipulate the control input device. In this example, the handle **4002** includes two grips that each include a finger loop **4004** and a grip member **4006** (grip members **4006***a* and **4006***b*). The two grip members **4006** are positioned on opposite sides of a central portion **4003** of the handle **4002**, where the grip members **4006** can be grasped, held, or otherwise contacted by a user's fingers. Each finger loop **4004** is attached to a respective grip member **4006** and can be used to secure a user's fingers to the associated grip member **4006**. Finger contacts **4005** can be connected or formed at the unconnected end of the grip members **4006***a* and **4006***b* to provide surfaces to contact the user's fingers.
(215) Each grip member **4006** and finger loop **4004** can be moved in an associated degree of freedom **4008** (e.g., **4008***a* and **4008***b*). In some examples, the grip members **4006***a* and **4006***b* are each coupled to the central portion **4003** of the handle **4002** at respective rotational couplings, allowing rotational movement of the grip members about grip axes **4007***a* and **4007***b*, respectively, with respect to the central portion **4003**. Each grip member **4006***a* and **4006***b* can be moved in an associated degree of freedom **4008***a* about axis **4007***a* and degree of freedom **4008***b* about axis **4007***b*, respectively, e.g., by a user contacting the grip members. For example, in some implementations the grip members **4006***a* and **4006***b* can be moved simultaneously in a pincher-type of movement (e.g., toward or away from each other). In various implementations, a single grip member **4006** and finger loop **4004** can be provided, or only one of the grip members **4006** can be moved in the degree of freedom **4008** while the other grip member **4006** can be fixed with reference to the handle **4002**. For example, the orientations of grip members **4006***a* and **4006***b* in their degrees of freedom can control corresponding rotational orientations of an end effector or other manipulator instrument. One or more grip sensors (not shown) can be coupled to the handle **4002** and/or other components of the device portion **4000** and can detect the orientations of the grip members **4006***a* and **4006***b* in their degrees of freedom **4008**. Some implementations can provide one or more active actuators (e.g., motors, voice coils, etc.) to output active forces on the grip members **4006** in the degrees of freedom **4008**, or passive actuators (e.g., brakes) or springs between the grip members **4006** and the central portion **4003** of the handle **4002** to provide resistance in directions of the grips.
(216) Handle **4002** is also provided with a rotational degree of freedom **4010** about a roll axis **4012** defined between a first end and second end of the handle **4002**. The roll axis **4012** is a longitudinal axis in this example that extends approximately along the center of the central portion **4003** of handle **4002**. For example, a user can rotate the grip members **4006** and central portion **4003** as a single unit around the axis **4012** with respect to a base of the controller portion **300**, such as housing **4009** to provide control of a manipulator system component. One or more control input sensors (not shown) can be coupled to the handle **4002** to detect the orientation of the handle **4002** in the rotational degree of freedom **4010**. Some implementations can provide one or more actuators to output forces on the handle **4002** in the rotational degree of freedom **4010**.
(217) Provided sensors can send signals describing sensed positions, orientations, and/or motions to one or more control circuits, e.g., a control system of the teleoperated system **100**. In some modes or implementations, the control circuits can provide control signals to a manipulator system, e.g., manipulator system **104** to, for example, control any of various degrees of freedom of an end effector or other instrument of the manipulator system **104**.

(218) In various implementations, the handle **4002** can be provided with additional degrees of freedom. For example, a rotational degree of freedom **4020** about a controller yaw axis **4022** can be provided to the handle **4002** at a rotational coupling between an elbow shaped link **4024** and a link **4026**, where the elbow shaped link **4024** is coupled to the handle **4002** (e.g., at housing **4009**). In this example, controller yaw axis **4022** intersects and is orthogonal to the roll axis **4012**. Additional degrees of freedom can similarly be provided. For example, link **4026** can be elbow-shaped and a rotational coupling can be provided between the other end of link **4026** and another link (not shown). A rotational degree of freedom **4028** about an axis **4030** can be provided to the handle **4002** at the rotational coupling. In some examples, the device portion **4000** can allow movement of the handle **4002** within a workspace of the user control system **102** with a plurality of degrees of freedom, e.g., six degrees of freedom including three rotational degrees of freedom and three translational degrees of freedom. One or more additional degrees of freedom can be sensed by control input sensors and/or actuated by actuators (motors, etc.) similarly as described above for the degrees of freedom **4008** and **4010**. In some implementations, each additional degree of freedom of the handle **4002** can control a different manipulator degree of freedom (or other motion) of an end effector of the manipulator system **104**.
(219) Various kinematic chains, linkages, gimbal mechanisms, flexible structures, or combinations of two or more of these can be used with the device portion **4000** in various implementations to provide one or more degrees of freedom to the controller portion. Some further examples of linkages and/or gimbal mechanisms that can be used with device portion **4000** are described in U.S. Pat. No. 6,714,839 B2, incorporated herein by reference.
(220) In the described example, handle **4002** includes one or more control switches **4050**, e.g., coupled to the central portion **4003**. In some implementations, the control switch **4050** can be moved to various positions to provide particular command signals, e.g., to select functions, options, or modes of the user control system and/or control input device (e.g., a controlling mode or non-controlling mode as described herein, or modes related to display system **300**, **600**, **700**, **2300**, or **3200**). In some implementations, control switch **4050** can be implemented as a button, a rotary dial, a switch, or other type of input control. Control switch **4050** can use optical sensors, mechanical switches, magnetic sensors, or other types of sensors to detect positions of the switch. In some implementations, handle **4002** also includes a presence sensing system including one or more presence sensors that can detect the presence of a user's hand operating the handle and/or the hand positioned near the handle.
(221) One or more display system features described herein can be used with other types of control input devices. For example, ungrounded control input devices can be used, which are free to move in space and disconnected from ground. In some examples, one or more handles similar to handle **4002** and/or grip members **4006** can be coupled to a mechanism worn on a user's hand and which is ungrounded, allowing the user to move grips freely in space. In some examples, the positions or orientations of the grips relative to each other and/or to other portions of the handle can be sensed by a mechanism coupling the grips together and constraining their motion relative to each other. Some implementations can use glove structures worn by a user's hand. Furthermore, some implementations can use sensors coupled to other structures to sense the grips within space, e.g., using video cameras or other sensors that can detect motion in **3**D space. Some examples of ungrounded control input devices are described in U.S. Pat. No. 8,543,240 B2 (filed Sep. 21, 2010) and U.S. Pat. No. 8,521,331 B2 (filed Nov. 13, 2008), both incorporated herein by reference in their entireties.
(222) FIG. **41** is a flow diagram illustrating an example method **4100** for operating a display system including one or more features described herein, according to some implementations. In some implementations, the method can, for example, be used with an example teleoperated system or other control system in which the display system provides a displayed view in conjunction with a teleoperated manipulator system. For example, in some implementations, the display system **300**

of FIG. **3**, display system **600** of FIG. **6**, display system **700** of FIG. **7**, display system **2300** of FIG. **23**, or display system **3200** of FIG. **32** can be used.
(223) In some examples, the display system can be used in a teleoperated system such as system **100** of FIG. **1**, where the display system is operated by a user who also operates one or more control input devices with his or her hand(s). A master-slave control relationship can be established between the control input device and a manipulator system, in which positions and orientations of the control input device are sensed and are transmitted to a control system, e.g., in the controller, manipulator system, and/or a separate control system. In some examples, motion of a control input device in space causes corresponding motion of a controlled instrument or other component of the manipulator system, and/or can control other functions of the manipulator system. The user input and control related to such teleoperation system components are not described in method **4100**.
(224) Other implementations of method **4100** can use a display system having one or more described features with other types of systems, e.g., non-teleoperated systems, a virtual environment implemented on a processing device and having no physical manipulator system and/or no physical subject interacting with a physical manipulator system, etc. In some implementations, the method can be partially or fully performed by a control circuit component, e.g., a control system. In some examples, the control circuit can include one or more processors, e.g., microprocessors or other control circuits, some examples of which are described below with reference to FIG. **42**.
(225) In block **4102**, a pivot axis is defined for the display unit of the display system, where the display unit provides a display of images for viewing by a user (e.g., display unit **306**, **606**, or **706** as described herein). The display unit is to be rotated about the defined pivot axis. In some cases or implementations, the defined pivot axis coincides with a rotational axis of a mechanical component of the display system, e.g., a tilt axis **326**, **626**, or **726** of a tilt member as described herein. In some implementations, the defined pivot axis is a virtual axis that is based on a combination of coordinated movements of the components of the display system and does not correspond to an axis of rotation of any particular mechanical component of the display system. For example, the virtual pivot axis can be a horizontal axis such as an eye pivot axis, a neck pivot axis, forehead pivot axis, or hand input device pivot axis as described herein in various implementations.
(226) In some implementations, the defined pivot axis can be changed to different locations in different modes or at different times of operation of the display system. In some implementations, the defined pivot axis is adjustable by the user. For example, the defined pivot axis can be determined based on user input received by the display system, e.g., via a user input device such as a keyboard, touchscreen, etc. in communication with the display system. In some examples, user input can specify one of multiple predetermined settings, where each setting defines the pivot axis at a different particular location in a workspace of the display unit of the display system. For example, one setting can define a particular eye pivot axis, and a different setting can define a particular neck pivot axis. In further examples, user input can adjust the location of the pivot axis. For example, the user input can specify to move a defined pivot axis a particular distance in space, along any of three dimensions. For example, user input can instruct a predefined eye pivot axis to be moved 5 millimeters in a vertical direction and 2 millimeters in a horizontal direction (e.g., toward a base support such as base support **302**).
(227) In block **4104**, the display unit is activated for operation. In some implementations, one or more presence sensors are provided on the display system to sense a presence of the user who is in a position to operate the display system, e.g., viewing the display of the display unit. The display unit can be activated and enabled to be moveable by user input in response to detecting user presence. For example, presence sensors can detect whether a user's head is positioned close to viewing ports or windows in the display unit. In some implementations, the display unit can be used in a teleoperated system such as system **100** of FIG. **1**, where the display unit is operated by a user who also operates one or more control input devices with his or her hand(s).

(228) In block **4106**, user input is received that directs the display unit to rotate and/or translate vertically, e.g., in a vertical plane. For example, user input can be received by user input devices of the display system, such as user input devices **340** and/or **342** on the display unit as described with reference to FIG. **3**. The user input may also be received from user input devices provided on other components of the display system, and/or from user input devices of a system that includes or is used in conjunction with the display system, e.g., a control input device of a teleoperated system. In one example, user input can instruct actuators of the vertical base support, horizontal arm support, and tilt mechanism to output forces that moves the display unit about the defined pivot axis. In some implementations, the user input can instruct to translate the display unit in its workspace. In some implementations, the user input instructs manipulation of images displayed by the display unit, and/or manipulation of other controlled devices, in accordance with the instructed movement of the display unit.
(229) In block **4108**, one or more actuators of the display system are controlled based on the user input to rotate and/or translate the display unit in a vertical plane. For example, the actuator(s) can be controlled to rotate the display unit in the vertical plane about the pivot axis determined in block **4102**, as described herein. In some implementations as described herein, a base support, arm support, and/or tilt member are moved by actuators simultaneously and in coordinated combination to cause the display unit to rotate about the defined pivot axis in accordance with the user input. In some implementations, the display unit can be translated linearly in a vertical plane based on the user input, e.g., moved based on linear movement provided by a base support and/or arm support.
(230) In some implementations, images or user interface controls displayed by the display unit are manipulated in accordance with the user input and/or in accordance with the movement of the display unit in the vertical plane. For example, the view of a displayed image can be moved in accordance with user input and/or motion of the display unit. In additional examples, user input can manipulate views or elements of a displayed user interface, virtual environment, or other display provided by the display system. For example, user interface features such as cursors, scroll bars, lists, menus, selections of graphical elements, etc., can be moved or manipulated in accordance with the user input and/or movement of the display unit. In some examples, such a user interface can be overlaid on a displayed captured view of a work site or can be displayed to the side(s) of the captured view.
(231) In some implementations, functions of a manipulator system can be manipulated in accordance with the user input and/or the display unit movement. For example, an actuator of a manipulator system can be controlled to move an image capture device (e.g., camera) of the manipulator system in correlation with the movement of the display unit, e.g., translating and/or rotating a camera based on corresponding motion of the display unit in its workspace. Image data is received from the image capture device and displayed by the display device of the display unit. The actuator is controlled based on commands sent to the manipulator system from the control system, where the commands are based on the user input and/or movement of the display unit. A different manipulator instrument can be controlled (e.g., via control input devices **210** and/or **212**) to move at least partially simultaneously with the movement of the manipulator image capture device controlled via the display unit, such that the user can control the manipulator image capture device to change the field of view of the slave work site displayed in display unit while the user also controls the different manipulator instrument to perform tasks in a teleoperated procedure at the manipulator work site as viewed via the display unit.
(232) Additional examples of teleoperated system functions manipulable by the user input can include, in some examples, a swap function causing control of a first manipulator arm or instrument to be swapped to a second manipulator arm or instrument; energy output for manipulator instruments; irrigation or suction at a surgical site; a clutch function to select a controlling mode and a non-controlling mode; etc.
(233) In block **4110**, yaw user input is received that directs the display unit to rotate about the yaw

axis (lateral rotation axis), e.g., left or right (lateral) rotation of the display unit about a vertical axis from the user's perspective. For example, the yaw user input can be received from similar user input devices as described for block **4106**, and/or from other user input devices of the display system or other system that are different from user input devices providing the user input of block **4106**. In various implementations, user input can instruct the display unit to rotate left about the yaw axis, to linearly translate the display unit left, to manipulate images or user interface elements displayed by the display unit, and/or to manipulate other controlled devices.

(234) In block **4112**, one or more actuators of the display system are controlled based on the user input to rotate the display unit about the yaw axis. For example, an actuator of the track mechanism described with respect to FIGS. **3-5**, FIG. **22**, FIGS. **23-25**, or FIGS. **32-38** can be used to rotate the display unit about yaw axis **330**, **630**, **730**, **2332**, or **3240**. In one example, user input that directs rotation to the left (e.g., counterclockwise as viewed from above) causes the actuator to output force that moves the display unit counterclockwise about the yaw axis. In some implementations, images or a user interface displayed by the display unit, and/or other controlled devices, are manipulated in accordance with the movement of the display unit similarly as described for block **4108**.

(235) Blocks **4106** and **4110** can be performed in any order and/or partially or completely simultaneously, such that, e.g., user input to rotate/translate the display unit in a vertical plane can be received simultaneously with user input to rotate the display unit about the yaw axis. Similarly, blocks **4108** and **4112** can be performed in any order and/or partially or completely simultaneously based on the user input received.

(236) The blocks **4106** to **4112** are repeated based on additional received user input that instructs the display unit to move vertically and/or about the yaw axis. Other user input may be received that deactivates the display unit operation, causes block **4102** to be performed to change the location of the defined pivot axis, or perform other functions of the display system.

(237) The blocks and operations described in the methods disclosed herein can be performed in a different order than shown and/or simultaneously (partially or completely) with other blocks and operations, where appropriate. Some blocks and operations can be performed for one portion of data and later performed again, e.g., for another portion of data. Not all of the described blocks and operations need be performed in various implementations. In some implementations, blocks and operations can be performed multiple times, in a different order, and/or at different times in the methods.

(238) Output such as haptic feedback on the display unit and/or visual output on a display device can be provided by the system to assist user operation of the display system and/or a teleoperated system. For example, a user interface may display warnings and/or error feedback on a display device, and/or audio output can be provided to indicate such warnings or errors. Such feedback can indicate, for example, reaching limits to the movement range of the display unit, moving the display unit such that a particular object being displayed is about to be moved out of view of the display unit, etc.

(239) In various implementations, other types of computer-assisted teleoperated systems, in addition to surgical systems, can be used with one or more display system features described herein. Such teleoperated systems can include controlled manipulator systems of various forms. For example, submersibles, hazardous material or device disposal units, industrial applications, applications in hostile environments and worksites (e.g., due to weather, temperature, pressure, radiation, or other conditions), general robotics applications, and/or remote-control applications (e.g., remote controlled vehicle or device with a first-person view), may utilize teleoperated systems that include manipulator or slave systems for sensory transmission (conveyed visual, auditory, etc. experience), manipulation of work pieces or other physical tasks, etc., and may use mechanically grounded and/or ungrounded control input devices to remotely control the manipulator systems and a display system to view worksites of the manipulator systems. Any such

teleoperated systems can be used with any of the various display system features described herein.
(240) FIG. **42** is a block diagram of an example master-slave system **4200**, which can be used with one or more implementations described herein.
(241) As shown, system **4200** includes a master device **4202** that a user may manipulate in order to control a slave device **4204** in communication with the master device **4202**. In some implementations, master device **1002** can be, or can be included in, any of user control systems described herein. In some implementations, slave device **1004** can be, or can be included in, manipulator system **104** of FIG. **1**. More generally, master device block **4202** can include one or more of various types of devices providing one or more control input devices that can be physically manipulated by a user. For example, master device **4202** can include a system of one or more master control devices such as one or more hand controllers (e.g., master control devices **210** and **212** or other hand controllers described herein).
(242) Master device **4202** generates control signals C**1** to Cx indicating positions and orientations, states, and/or changes of one or more controllers in their degrees of freedom. For example, the master device **4202** can generate control signals indicating selection of input controls such as physical buttons, hand controller states, and other manipulations of the hand controller by the user. The control signals can be sent via wired connections, or wireless.
(243) A control system **4210** can be included in the master device **4202**, in the slave device **4204**, or in a separate device, e.g., an intermediary device communicatively connected between master device **4202** and slave device **4204**. In some implementations, the control system **4210** can be distributed among multiple of these devices.
(244) In the example system, control system **4210** receives control signals C**1** to Cx and generates actuation signals A**1** to Ay, which are sent to slave device **4204**. Control system **4210** can also receive sensor signals B**1** to By from the slave device **4204** that indicate positions and orientations, states, and/or changes of various slave components (e.g., manipulator arm elements). These actuation and/or sensor signals can be sent via wired connections, or wireless.
(245) Control system **4210** can include components such as a processor **4212**, memory **4214**, and interface hardware **4216** and **4218** such as a master interface and a slave interface for communication with master device **4202** and slave device **4204**, respectively. Processor **4212** can execute program code and control operations of the system **4200**, including operations related to processing sensor data from a display unit and commanding modes and components as described herein. Processor **4212** can include one or more processors of various types, including microprocessors, application specific integrated circuits (ASICs), and other electronic circuits. Memory **4214** can store instructions for execution by the processor and can include any suitable processor-readable storage medium, e.g., random access memory (RAM), read-only memory (ROM), Electrical Erasable Read-only Memory (EEPROM), Flash memory, etc. For example, the processor **4212** and memory **4214** enable control between the master control device **4202** and the slave device **4204**.
(246) The control system **4210** includes programmed instructions (e.g., a computer-readable medium storing the instructions) to implement appropriate operations and blocks of methods in accordance with aspects disclosed herein. The system may include multiple data processing circuits with one portion of the processing optionally being performed on or adjacent the slave device **4204**, another portion of the processing being performed at the master device **4202**, etc. Any of a wide variety of centralized or distributed data processing architectures may be employed. Similarly, the programmed instructions may be implemented as a number of separate programs or subroutines, or they may be integrated into a number of other aspects of the teleoperated systems described herein. In one embodiment, control system **4210** supports one or more wireless communication protocols such as Bluetooth, IrDA, HomeRF, IEEE 802.11, DECT, and Wireless Telemetry.
(247) Various other input and output devices can also be coupled to the control system **4210**. The output devices can include one or more display units as a component of a display system **4220**. For

example, display system **4220** can be similar to display system **300**, **600**, **700**, or **2300** described herein. Display system **4220** receives command and/or data signals E from the control system **4210** to cause display of images, e.g., digital images composed of multiple pixels. The images can be generated by the control system or other system (e.g., graphical elements and animations in a user interface allowing selection of commands for a teleoperated system), and/or the images can be derived from data captured by an image capture device positioned at a slave device or other location. Display system **4220** also provides command signals F**1** to Fx to the control system **4210** to control functions of the display system, including movement of a display unit (e.g., display unit **306**, **606**, **706**, or **2306**) in space as described herein. For example, the command signals F**1** to Fx can be generated based on user manipulation of user input devices, as described herein, and/or based on detected events or other conditions. In some implementations, display system **4220** can enable or disable (e.g., follow or ignore) the commands F**1** to Fx based on the state of the teleoperated system or control system **4210**.

(248) In this example, control system **4210** includes a mode control module **4240**, a controlling mode module **4250**, and a non-controlling mode module **4260**. Other implementations can use other modules, e.g., a force output control module, sensor input signal module, etc. As used herein, the term “module” can refer to a combination of hardware (e.g., a processor such as an integrated circuit or other circuitry) and software (e.g., machine or processor executable instructions, commands, or code such as firmware, programming, or object code). A combination of hardware and software can include hardware only (i.e., a hardware element with no software elements), software hosted by hardware (e.g., software that is stored at a memory and executed or interpreted by or at a processor), or a combination of hardware and software hosted at hardware. In some implementations, the modules **4240**, **4250**, and **4260** can be implemented using the processor **4212** and memory **4214**, e.g., program instructions stored in memory **4214** and/or other memory or storage devices connected to control system **4210**.

(249) Mode control module **4240** can detect when a user initiates a controlling mode and a non-controlling mode of the system, e.g., by user selection of controls, sensing a presence of a user using a master controller, sensing required manipulation of a master controller, etc. The mode control module can set the controlling mode or a non-controlling mode of the control system **4210** based on one or more control signals C**1** to Cx. For example, mode control module **4240** may activate controlling mode operation if user detection module detects that a user is in proper position for use of the control input device(s) and that signals (e.g., one or more signals C**1** to Cx) indicate the user has contacted the control input device(s). The mode control module **4240** may disable controlling mode if no user touch is detected on the control input device(s) and/or if a user is not in proper position for use of the control input device(s). For example, the mode control module **4240** can inform control system **4210** or send information directly to controlling mode module **4250** to prevent the controlling mode module **4250** from generating actuation signals A**1** to An that move slave device **4204**.

(250) In some implementations, controlling mode module **4250** may be used to control a controlling mode of control system **4210**. Controlling mode module **4250** can receive control signals C**1** to Cx and can generate actuation signals A**1** to Ay that control actuators of the slave device **4204** and cause it to follow the movement of master device **4202**, e.g., so that the movements of slave device **4204** correspond to a mapping of the movements of master device **4202**. Controlling mode module **4250** can be implemented using conventional techniques. In some implementations, controlling mode module **4250** can also be used to control forces on the master device **4202**. For example, forces can be output on one or more components of the master control devices, e.g., hand grip members or mechanical links of the control input devices, using one or more control signals D**1** to Dx output to actuator(s) used to apply forces to the components. Such forces can provide haptic feedback, gravity compensation, etc.

(251) In some implementations, a non-controlling mode module **4260** may be used to control a

non-controlling mode of system **4200**. In the non-controlling mode, user manipulations of master device **4202** have no effect on the movement of one or more components of slave **4204**. In some examples, non-controlling mode may be used when a portion of slave device **4204**, e.g., a manipulator arm assembly, is not being controlled by master device **4202**, but rather is floating in space and may be manually moved. For non-controlling mode, non-controlling mode module **4260** may allow actuator systems in the slave device **4204** to be freewheeling or may generate actuation signals A**1** to An, for example, to allow motors in a manipulator arm to support the expected weight of the arm against gravity, where brakes in the arm are not engaged and permit manual movement of the arm.

(252) In some implementations, non-controlling mode can include one or more other operating modes of the control system **4210**. For example, a non-controlling mode can be a selection mode in which movement of a control input device in one or more of its degrees of freedom and/or selection of controls of the control input device can control selection of displayed options, e.g., in a graphical user interface displayed by display system **4220** and/or other display devices. A viewing mode can allow movement of control input devices to control a display provided from imaging devices (e.g., cameras), or movement of imaging devices, that may not be included in the slave device **4204**. Control signals C**1** to Cx can be used by the non-controlling mode module **4260** to control such elements (e.g., cursor, views, etc.) and control signals D**1** to Dx can be determined by the non-controlling mode module to cause output of forces on the control input device(s) during such non-controlling modes, e.g., to indicate to the user interactions or events occurring during such modes.

(253) Implementations described herein may be implemented, at least in part, by computer program instructions or code, which can be executed on a computer. For example, the code may be implemented by one or more digital processors (e.g., microprocessors or other processing circuitry). Instructions can be stored on a computer program product including a non-transitory computer readable medium (e.g., storage medium), where the computer readable medium can include a magnetic, optical, electromagnetic, or semiconductor storage medium including semiconductor or solid state memory, magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), flash memory, a rigid magnetic disk, an optical disk, a memory card, a solid-state memory drive, etc. The media may be or be included in a server or other device connected to a network such as the Internet that provides for the downloading of data and executable instructions. Alternatively, implementations can be in hardware (logic gates, etc.), or in a combination of hardware and software. Example hardware can be programmable processors (e.g. Field-Programmable Gate Array (FPGA), Complex Programmable Logic Device), general-purpose processors, graphics processors, Application Specific Integrated Circuits (ASICs), and the like.

(254) Note that the functional blocks, operations, features, methods, devices, and systems described in the present disclosure may be integrated or divided into different combinations of systems, devices, and functional blocks as would be known to those skilled in the art.

(255) Although the present implementations have been described in accordance with the examples shown, one of ordinary skill in the art will readily recognize that there can be variations to the implementations and those variations would be within the scope of the present disclosure. Accordingly, many modifications may be made by one of ordinary skill in the art without departing from the scope of the appended claims.

Claims

1. A control unit comprising: a first support; a second support coupled to the first support, wherein the second support is linearly translatable along a first axis in a first degree of freedom with respect to the first support, and at least a portion of the second support is linearly translatable along a second axis in a second degree of freedom with respect to the first support; a display unit rotatably

coupled to the second support, wherein the display unit is rotatable about a third axis in a third degree of freedom with respect to the second support; and a first actuator, a second actuator, and a third actuator, wherein the first actuator is configured to output first forces on the display unit in the first degree of freedom, the second actuator is configured to output second forces on the display unit in the second degree of freedom, and the third actuator is configured to output third forces on the display unit in the third degree of freedom, wherein the display unit is rotatable about a defined pivot axis based on a coordinated combination of linear translation of the second support in the first degree of freedom, linear translation of at least the portion of the second support in the second degree of freedom, and rotation of the display unit in the third degree of freedom, wherein the first actuator, the second actuator, and the third actuator are configured to output the first forces, the second forces, and the third forces in combination to cause rotation of the display unit about the defined pivot axis, wherein the defined pivot axis does not correspond to an axis of rotation of any mechanical joint of the control unit, and wherein the defined pivot axis is configured to intersect a portion of a user operating the display unit or intersect a portion of the display unit.
2. The control unit of claim 1, wherein: the first support includes a first telescoping base portion and a second telescoping base portion, the second telescoping base portion is linearly translatable along the first axis with respect to the first telescoping base portion, the second support includes a first telescoping arm portion and a second telescoping arm portion, the second telescoping arm portion is linearly translatable along the second axis with respect to the first telescoping arm portion, and the second telescoping base portion of the first support is rigidly coupled to the first telescoping arm portion of the second support.
3. The control unit of claim 1, wherein: the second support is coupled to the first support by a middle support, the middle support includes a horizontal portion coupled rigidly to a vertical portion, the horizontal portion and the vertical portion being orthogonal to each other, the second support is horizontally translatable in the second degree of freedom with respect to the middle support, and the middle support and the second support are vertically translatable in the first degree of freedom with respect to the first support.
4. The control unit of claim 1, wherein: the display unit includes a tilt member rotatably coupled to an end of the second support, the tilt member rotatable about the third axis in the third degree of freedom, and the third axis is orthogonal to the first axis and to the second axis.
5. The control unit of claim 4, wherein: the display unit is rotatable about a fourth axis with respect to the tilt member in a fourth degree of freedom, and the fourth axis is orthogonal to the third axis.
6. The control unit of claim 4, further comprising: an actuator configured to output a force on the display unit about a fourth axis in a fourth degree of freedom; and a curved track coupled to the tilt member, wherein the display unit is coupled to the curved track, wherein the display unit, the curved track, and the tilt member are rotatable about the third axis in the third degree of freedom with respect to the second support, wherein the display unit is guided by the curved track to be rotatable with respect to the tilt member about the fourth axis in the fourth degree of freedom, and wherein the fourth axis is a yaw axis.
7. The control unit of claim 1, wherein: the portion of the second support includes a yoke portion including two yoke members, and the display unit is rotatably coupled to the two yoke members and is positioned between the two yoke members.
8. The control unit of claim 4, wherein a first end of the tilt member is coupled to an end of the second support, wherein the tilt member rotates within a rotatable range in which the tilt member linearly extends from the first end of the tilt member toward the first support to a second end of the tilt member, wherein the display unit is coupled to the second end of the tilt member.
9. The control unit of claim 1, wherein the first actuator, the second actuator, and the third actuator are configured to output the first forces, the second forces, and the third forces in combination to cause rotation of the display unit about the defined pivot axis to follow movement of a head of a user that operates the control unit.

10. The control unit of claim 1, wherein the display unit includes: a hand input device provided on the display unit and configured to receive hand input from a hand of the user, wherein the hand input includes commands that cause the hand input device to output control signals to cause the actuators to move the display unit in a workspace of the display unit about the defined pivot axis.
11. The control unit of claim 1, wherein the control unit is coupled to a device including a control input device manipulable by a user to control one or more functions of a teleoperated manipulator system, wherein the one or more functions include causing an actuator of the teleoperated manipulator system to move an image capture device of the teleoperated manipulator system in accordance with manipulation of the control input device, wherein image data is received from the image capture device and displayed by the display unit.
12. The control unit of claim 1, wherein the display unit includes a head input device provided on the display unit and configured to receive head input via contact of a head of the user with the head input device, wherein the head input includes commands to move the display unit in a workspace of the display unit about the defined pivot axis.
13. The control unit of claim 1, wherein: the defined pivot axis is positioned at a location such that the defined pivot axis extends through a neck of a user when the user operates the display unit; the defined pivot axis extends through a hand input device provided on the display unit; the defined pivot axis extends through a head input device provided on the display unit at a point on the head input device that is configured to contact a forehead of a user that operates the head input device; or the defined pivot axis is coincident with an eye axis extending through the eyes of the user that operates the display unit.
14. The control unit of claim 1, wherein: the defined pivot axis is configured to intersect a portion of the display unit.
15. A control unit comprising: a support mechanism including: a support linkage including a plurality of links including a first support and a second support; a display unit coupled to the support linkage, wherein the display unit is moveable in multiple degrees of freedom based on relative movement between the plurality of links, wherein the multiple degrees of freedom include a linear first degree of freedom, a linear second degree of freedom, and a rotational third degree of freedom, wherein the display unit is rotatable about a defined pivot axis based on a coordinated combination of movement including linear translation of the second support in the first degree of freedom, linear translation of at least a portion of the second support in the second degree of freedom, and rotation of the display unit in the third degree of freedom; and a plurality of actuators coupled to the support linkage; and a control system in communication with the support mechanism and configured to provide control signals to the plurality of actuators to cause the display unit to rotate about a defined pivot axis, wherein the rotation about the defined pivot axis results from a coordinated combination of movement of the display unit in the linear first degree of freedom, the linear second degree of freedom, and the rotational third degree of freedom, wherein the defined pivot axis does not correspond to an axis of rotation of any mechanical joint of the control unit, and wherein the defined pivot axis is configured to intersect a portion of a user operating the display unit or intersect a portion of the display unit.
16. The control unit of claim 15, wherein the defined pivot axis is a horizontal axis aligned with an axis extending through a portion of a hand input device provided on the display unit, wherein the hand input device is configured to be operated by a hand of the user operating the display unit and the hand input device, wherein the hand input device is operable to receive hand input from the hand of the user, wherein the hand input includes commands that cause the hand input device to output control signals to cause the actuators to move the display unit in a workspace of the display unit about the defined pivot axis.
17. The control unit of claim 15, wherein: the defined pivot axis is a horizontal axis extending through a neck of a user that operates the display unit; the defined pivot axis is a horizontal axis aligned with an axis extending through the eyes of the user that operates the display unit; or the

defined pivot axis is a horizontal axis aligned with an axis extending through a portion of an input device provided on the display unit, wherein the portion of the input device is configured to contact a forehead of the user that operates the display unit.
18. The control unit of claim 15, wherein the defined pivot axis is adjustable in space based on user input to the control unit that defines a location of the defined pivot axis.
19. The control unit of claim 15, wherein the control system is configured to provide the control signals to control the actuators to move the display unit about the defined pivot axis to follow movement of a head of a user that operates the control unit.
20. A method comprising: determining a defined pivot axis for a display unit; receiving first user input at a first input device; and causing movement of the display unit in one or more degrees of freedom provided by a support linkage coupled to the display unit, wherein the movement is based on the first user input, wherein causing the movement includes: controlling a first actuator to cause a second link of the support linkage to linearly translate with respect to a first link of the support linkage along a first axis in a first degree of freedom; controlling a second actuator to cause a portion of the second link to linearly translate with respect to the first link along a second axis in a second degree of freedom; and controlling a third actuator to cause the display unit to rotate about a third axis in a third degree of freedom with respect to the second link, wherein the first actuator, the second actuator, and the third actuator are controlled to cause a coordinated combination of linear translation in the first degree of freedom, linear translation in the second degree of freedom, and rotation of the display unit in the third degree of freedom to cause the display unit to rotate about the defined pivot axis, wherein the defined pivot axis does not correspond to an axis of rotation of any mechanical joint of the support linkage and the display unit, and wherein the defined pivot axis is configured to intersect a portion of a user operating the display unit or intersect a portion of the display unit.